

Differentially Private Explanations for Clusters

Abstract

The dire need to protect sensitive data has led to various flavors of privacy definitions. Among these, Differential privacy (DP) is considered one of the most rigorous and secure notions of privacy, enabling data analysis while preserving the privacy of data contributors. One of the fundamental tasks of data analysis is clustering, which is meant to unravel hidden patterns within complex datasets. However, interpreting clustering results poses significant challenges, and often necessitates an extensive analytical process. Interpreting clustering results under DP is even more challenging, as analysts are provided with noisy responses to queries, and longer, manual exploration sessions require additional noise to meet privacy constraints. While increasing attention has been given to clustering explanation frameworks that aim at assisting analysts by automatically uncovering the characteristics of each cluster, such frameworks may also disclose sensitive information within the dataset, leading to a breach in privacy.

To address these challenges, we present DPCLustX, a framework that provides explanations for black-box clustering results while satisfying DP. DPCLustX takes as input the sensitive dataset alongside privately computed clustering labels, and outputs a global explanation, emphasizing prominent characteristics of each cluster while guaranteeing DP. We perform an extensive experimental analysis of DPCLustX on real data, showing that it provides insightful and accurate explanations even under tight privacy constraints.

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1 Introduction

Sensitive data collection has never been more prevalent, from fitness trackers [20] and financial institutions [50], to healthcare and insurance facilities [2]. Such data require special treatment to allow for its safe and secure usage without exposing individuals to harm due to the presence of their information in the data. Differential Privacy (DP) [14, 17] has emerged as the gold standard for handling such issues. The crux of DP is to allow sensitive data to be used while bounding the privacy leakage and to offer utility bounds on the obtained results. Indeed, DP has been adopted by multiple companies [3, 15] and governmental organizations [10, 18, 58].

Among the many operations in data analysis, *clustering* plays a pivotal role in uncovering hidden patterns and providing actionable insights from data. To ensure privacy, researchers have worked extensively to adapt clustering techniques to comply with DP [23, 25, 55, 56]. Under DP constraints, the true clusters must be obfuscated and slightly distorted to prevent individual information leakage, often at the cost of accuracy. Such clustering algorithms are inherently complex even without privacy considerations, and the additional requirement to adhere to strict DP standards further amplifies this complexity.

To account for this, previous work has focused on *explaining non-private clustering algorithms* [19, 21, 41, 47, 60] and aimed to provide succinct and interpretable summaries of the properties of each cluster by showing how it varies from the other clusters. When considering privacy, it is likely that lack of access to the data is accompanied by lack of access to the clustering algorithm, requiring an approach that is suitable for black-box clustering algorithms. Works that focus on providing explanations for black-box clustering methods are usually histogram-based approaches [7, 9], which is a popular form of explanation in other settings as well [35, 38, 57, 63]. With these approaches, users get the clusters along with a *histogram for each cluster* that focuses on a specific attribute and graphically shows how the data associated with the cluster differs from the rest of the data. Yet, the approaches that generate such explanations cannot be directly used in the DP setting. In particular, these approaches choose histograms based on quality functions, such as interestingness [7, 9, 28, 54], sufficiency [7, 8], and diversity [7, 66] which would have to be significantly distorted to satisfy DP, making it impractical to obtain a reliable explanation. Moreover, existing methods generate all explanation options and then choose the ones with the highest scores. However, in the DP setting, this strategy quickly becomes infeasible because it requires producing private histograms for every attribute and cluster, necessitating excessive distortion to ensure DP compliance.

In light of this, we propose DPCLustX, a novel framework for generating histogram-based explanations of clustering results under DP for black-box clustering methods. DPCLustX is inspired by previous work [7, 9] and addresses the above limitations as follows. (1) We first prove that previous approaches cannot be applied directly, as existing quality functions for histogram-based explanations would have to be significantly distorted to satisfy DP. Then, we develop DP tailored variants that enable the generation of high-quality and privacy-preserving explanations. (2) To minimize privacy costs in the explanation selection process, we evaluate the *attribute* quality directly, privately selecting high-scoring explaining attributes based on the sensitive dataset with our novel quality function. We then generate noisy histograms exclusively for the selected attributes, leveraging previous work on DP-preserving noisy histograms [11, 16]. Additionally, recognizing that the search space for the best clustering explanation is large, (3) we adapt techniques from prior work [7] and develop a DP mechanism that constructs a *candidate set* for each cluster, from which the clustering explanation is privately derived. To this end, we adapt the idea of a

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age	gender	lab_proc	diag_1	...	cluster
[60, 70)	Female	[40, 50)	Circulatory	...	1
[60, 70)	Female	[0, 10)	Diabetes	...	2
[70, 80)	Male	[40, 50)	Injury	...	2
...	...	...	...	...	...

Figure 1: Example of the Diabetes dataset.

single-cluster score function which is used to rank the attributes for each cluster by their explanation quality for that cluster [7] and tailor it to the DP setting. This function is adapted to DP so that the noise added to it will still render the results useful in filtering attributes. Then, we privately select Top- k explaining attributes for each cluster, constructing a high-quality candidate pool. However, since iteratively applying the exponential mechanism [43] k times for each cluster can be computationally expensive, we utilize the One-shot Top- k mechanism [13] to privately select top- k , ensuring fast execution times for interactive user experience. An illustration of the DP-ClustX framework, summarizing these steps, is given in Figure 3. Our experimental study confirms that DP-ClustX generates insightful and accurate explanations even under tight privacy constraints, while maintaining reasonable execution times.

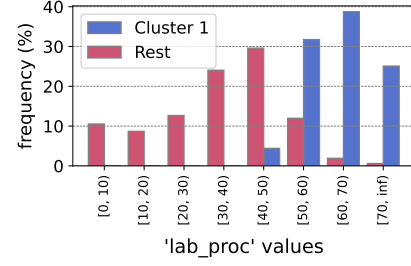
Example 1.1. Consider an analyst working with the Diabetes dataset [6] (a subset of the columns is illustrated in Figure 1), aiming to identify groups of patients with similar medical records using the DP- k -means algorithm [56]. While DP- k -means privately provides cluster centers (see the last column in Figure 1 associating each tuple with a cluster), it does not offer additional insights about the clusters. Instead of exhausting the privacy budget through a manual EDA session, the analyst employs DP-ClustX to generate histogram-based explanations. These explanations reveal significant patterns, such as the fact that Cluster 1 consists primarily of individuals who underwent a higher number of lab procedures, as shown in Figure 2a. Comparing the cluster distribution of lab_proc with the remaining data, we see that values outside Cluster 1 are concentrated in the middle and lower ranges, peaking at [40, 50). In contrast, Cluster 1 values differ greatly, being concentrated in the upper range, peaking at [60, 70). This suggests that the clustering algorithm groups individuals with a higher number of lab procedures in Cluster 1. For simplicity, we have attached an LLM generated textual description of the histogram in Figure 2b.

Our contributions. Our contributions are summarized as follows:

- We formulate the problem of finding the optimal privacy-preserving histogram-based explanation for black-box clustering methods (Section 3).
- We design quality functions for histogram-based explanations inspired by the notions of interestingness, sufficiency, and diversity that are adapted for the DP setting (Section 4).
- We develop DP-ClustX, the first framework designed to generate histogram-based explanations for clustering results under DP (Section 5), equipped with formal guarantees.
- We provide a comprehensive experimental study demonstrating the effectiveness of DP-ClustX, showing that its explanations align closely with non-private explanations even under a strict privacy budget (Section 6).

2 Preliminaries

In this section, we introduce our notations, and review existing concepts from DP used in this work.



(a) Part of a histogram-based explanation for the Diabetes dataset. The selected attribute, lab_proc, specifies the number of lab procedures an individual underwent.

Textual description: The 'lab_proc' column values differ significantly. Values outside Cluster 1 are concentrated in the lower and mid-range (85% below 50), while Cluster 1 contains mainly higher values (95% above 50).

(b) Textual description of the histogram explanation in Figure 2a.

Figure 2: Cluster explanation with its textual description.

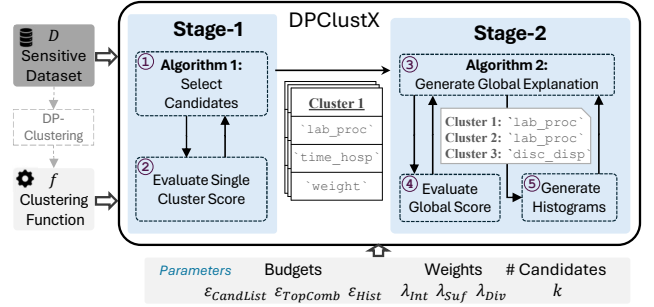


Figure 3: System architecture of DP-ClustX. Algorithm 1 selects candidate attributes for each cluster (1) using the single-cluster score function (2). Algorithm 2 selects a high-quality attribute combination from these candidates (3) using the global score function (4), and generates noisy histograms only for the selected attributes (5).

Data. Given a single table schema $S = R(A_1, \dots, A_d)$, R is a relation name and $\mathcal{A} = \{A_1, \dots, A_d\}$ denotes the set of attributes. Each attribute A_i has discrete, finite, and data-independent domain $\text{dom}(A_i)$. The full domain of R is $\text{dom}(R) = \text{dom}(A_1) \times \dots \times \text{dom}(A_d)$. An instance (dataset) D of a relation R is a bag (multiset) whose elements are tuples in $\text{dom}(R)$. We let $\mathcal{D}$ denote the set of all datasets of the relation R , i.e., $\mathcal{D} = \{D \mid D \subseteq \text{dom}(A_1) \times \dots \times \text{dom}(A_d), |D| < \infty\}$. The number of tuples in D is denoted as $|D|$. $\pi_A(D)$ is the projection of D onto the attribute A . For a dataset D and an attribute $A \in \mathcal{A}$, we let $\text{dom}_D(A)$ denote the *active domain* of A with respect to D , i.e., the subset of $\text{dom}(A)$ of values appearing in $\pi_A(D)$ at least once.

Example 2.1. Consider the Diabetes dataset, illustrated in Figure 1, which contains 47 attributes, including age, gender, lab_proc, and diag_1. The domain of lab_proc, $\text{dom}(\text{lab_proc})$, comprises 8 values, each representing a range of lab procedure counts. Let D_1

be the cluster explained in Example 1.1. Here, $\text{dom}_{D_1}(\text{lab_proc})$ comprises 7 values, as no tuple in D_1 has $\text{lab_proc} = [20, 30]$.

Histograms. For a dataset D and an attribute A , we let $\text{cnt}_{A=a}(D)$ denote the number of occurrences of value a in $\pi_A(D)$. Let $h_A(D) \in (\mathbb{N} \cup \{0\})^{\text{dom}(A)}$ denote the histogram of the dataset D over attribute A . That is, $h_A(D)[a] = \text{cnt}_{A=a}(D)$ for any $a \in \text{dom}(A)$. For visualizations, we use normalized histograms, where each count is replaced by its proportion.

Example 2.2. Figure 2a illustrates two histograms derived from the Diabetes dataset. For instance, the orange bars represent the histogram $h_{\text{lab_proc}}(D_1)$.

Histogram-based explanation (HBE). A clustering of D is a partition into disjoint subsets $\{D_c \subseteq D \mid c \in C\}$ each assigned with a cluster label $c \in C$. That is, $D_c \cap D_{c'} = \emptyset$ for $c \neq c'$ and $\bigcup_{c \in C} D_c = D$. An explanation candidate for a single cluster D_c consists of two histograms on a specified attribute: the histogram of the cluster values, and the histogram of the values outside the cluster (illustrated in Figure 2a). A global explanation candidate is a set of single cluster explanation candidates, with a candidate for each cluster. Formally,

Definition 2.3 (Single cluster HBE candidate [7, 9]). For a dataset D and a cluster label $c \in C$, an explanation candidate e_c is a tuple $(c, A, h_A(D \setminus D_c), h_A(D_c))$ where $A \in \mathcal{A}$.

Example 2.4. Figure 2a illustrates single a cluster HBE candidate for the dataset Diabetes. In this case, the attribute is lab_proc , which represents the number of lab procedures an individual underwent. The (normalized) histogram of the cluster values is shown in blue, and for the remaining data in red.

Definition 2.5 (Global HBE candidate [7]). Given D , clustered into $\{D_c \mid c \in C\}$, a global HBE candidate is a set $\{e_c \mid c \in C\}$, where e_c is a single-cluster explanation candidate for D_c .

Assessing the quality of HBEs. An HBE should capture unique patterns that characterize each cluster, and distinguish it from other clusters. Building upon [8, 9, 28, 66], the work of [7] proposed the aspects of *interestingness*, *sufficiency* and *diversity* for evaluating the quality of an HBE. Briefly, interestingness is quantified as the distributional shift between the inner cluster distribution and the full dataset in a given attribute. Sufficiency represents the extent to which the HBE captures the patterns of the underlying clustering, and diversity measures the overall distinctiveness among explanations. We discuss these measures further in Section 4.

2.1 Differential Privacy

We next review preliminary background from the DP literature used in this work. DP ensures that the distribution of outputs does not significantly change when the algorithm is applied to any two *neighboring databases*.

Definition 2.6 (Neighboring Datasets [14]). Two datasets D and D' are called *neighboring* (denoted $D \sim D'$) if D' can be obtained from D by removing or adding one tuple, i.e., $|D \setminus D'| \cup (D' \setminus D)| = 1$.

Definition 2.7 (ϵ -Differential Privacy (DP) [14, 17]). A randomized mechanism $\mathcal{M}$ is said to satisfy ϵ -DP if for any neighboring datasets $D \sim D'$ and any set of possible outputs $S \subseteq \text{Range}(\mathcal{M})$,

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \cdot \Pr[\mathcal{M}(D') \in S],$$

where the probability is over the randomness of $\mathcal{M}$.

PROPOSITION 2.8. *The following holds for DP [16, 17]:*

- *Sequential Composition:* Let $\mathcal{M}_1 : \mathcal{D} \rightarrow \mathcal{Y}$ and $\mathcal{M}_2 : \mathcal{D} \times \mathcal{Y} \rightarrow \mathcal{Z}$. Suppose $\mathcal{M}_1$ satisfies ϵ_1 -DP and for every $y \in \mathcal{Y}$, $\mathcal{M}_2(\cdot, y)$ satisfies ϵ_2 -DP (as a function of its first input). Define $\mathcal{M}_3 : \mathcal{D} \rightarrow \mathcal{Z}$ by $\mathcal{M}_3(D) = \mathcal{M}_2(D, \mathcal{M}_1(D))$. Then, $\mathcal{M}_3$ satisfies $(\epsilon_1 + \epsilon_2)$ -DP.
- *Parallel Composition:* Suppose $\mathcal{M}_1 : \mathcal{D} \rightarrow \mathcal{Y}$ satisfies ϵ_1 -DP and $\mathcal{M}_2 : \mathcal{D} \rightarrow \mathcal{Z}$ satisfies ϵ_2 -DP. Let $\mathcal{D}_1, \mathcal{D}_2 \subseteq \text{dom}(R)$ be disjoint subsets of the input domain. Define $\mathcal{M}' : \mathcal{D} \rightarrow \mathcal{Y} \times \mathcal{Z}$ by $\mathcal{M}'(D) = (\mathcal{M}_1(D \cap \mathcal{D}_1), \mathcal{M}_2(D \cap \mathcal{D}_2))$. Then, $\mathcal{M}'$ satisfies $\max\{\epsilon_1, \epsilon_2\}$ -DP.
- *Post-processing:* Suppose $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{Y}$ satisfies ϵ -DP. Then, for any function $g : \mathcal{Y} \rightarrow \mathcal{Z}$ (deterministic or randomized), The mechanism defined by $g(\mathcal{M}(D))$ satisfies ϵ -DP.

To quantify the noise that has to be injected in order for mechanisms to satisfy DP, we first define then notion of sensitivity,

Definition 2.9 (Sensitivity [17]). For a set of candidates $\mathcal{R}$, Let $q : \mathcal{D} \times \mathcal{R} \rightarrow \mathbb{R}$ be a quality function that given $D \in \mathcal{D}$, defines a score for every $r \in \mathcal{R}$. The sensitivity of q is

$$\Delta_q = \sup_{r \in \mathcal{R}} \sup_{D \sim D'} |q(D) - q(D')|$$

The exponential mechanism [43] (Definition 2.10) is a DP primitive for privately releasing the top item from a set of candidates with respect to a quality function that depends on the sensitive dataset, with utility and privacy guarantees (Theorem 2.11). The mechanism injects noise to the selection process, and outputs each candidate with probability proportional to its score.

Definition 2.10 (The Exponential Mechanism (EM) [43]). Given $D \in \mathcal{D}$, a set of candidates $\mathcal{R}$, a quality function $q : \mathcal{D} \times \mathcal{R} \rightarrow \mathbb{R}$, and a privacy parameter ϵ , the exponential mechanism $\mathcal{M}_E$ selects and outputs $r \in \mathcal{R}$ with probability proportional to $\exp\left(\frac{\epsilon \cdot q(D, r)}{2\Delta_q}\right)$.

THEOREM 2.11 (EM PRIVACY [43]). *The exponential mechanism satisfies ϵ -DP.*

Differentially private histograms. DP mechanisms for computing private histograms are well-studied (e.g., [4, 27, 36, 52, 65]). Since our framework can be instantiated with any DP histogram generation mechanism, we denote it as $\mathcal{M}_{\text{hist}}(\pi_A(D), \epsilon_{\text{hist}})$. It takes the column of interest $\pi_A(D)$ and a privacy budget ϵ_{hist} as inputs, and outputs a histogram of noisy counts $\tilde{h}_A(D)$ over $\text{dom}(A)$, while satisfying ϵ_{hist} -DP. Such mechanisms are accompanied by utility bounds, enabling analysts to control histogram accuracy by translating accuracy requirements into the required privacy budget.

Differentially private clustering. DP clustering has been extensively studied in the DP literature, with prominent approaches aiming to release cluster centers from a sensitive dataset D while

preserving differential privacy (e.g., [23, 25, 55, 56]). In the non-private black-box clustering explanation setting (e.g. [60]), a clustering is typically modeled by a function $f : D \rightarrow C$, assigning each tuple in D a cluster label $c \in C$. However, this modeling is inherently unsuitable for the output of DP clustering, which requires any possible output (i.e., a clustering function) to occur with similar probability for any two different but neighboring datasets. Instead, we model the output of a DP clustering algorithm as function $f : \text{dom}(R) \rightarrow C$. This definition encompasses DP mechanisms that release centers, as the fixed centers define a cluster assignment for any tuple in $\text{dom}(R)$, while also accommodating other approaches, such as user-defined predicates or future clustering techniques.

3 Problem Statement

We begin by describing the privacy setup, which is inspired by previous work on DP black-box classifier explanations [51]. Subsequently, we present the formal problem definitions.

An HBE mechanism (Definition 3.1) preserves privacy if for any clustering function $f : \text{dom}(R) \rightarrow C$, its distribution of outputs does not change much when we add a single tuple to the dataset. Note that we assume only *black-box* access to f , and therefore make no assumptions regarding the formation or structure of the clusters.

Definition 3.1. An HBE mechanism $\text{Exp}(D, f)$ is an algorithm that takes a dataset D and a clustering function $f : \text{dom}(R) \rightarrow C$, and outputs a global HBE (Definition 2.5) for the clustering of D defined by f . We say that Exp satisfies ϵ_{exp} -DP if for any clustering function f , for any pair of neighboring datasets $D \sim D'$, and any set of explanations $S \subseteq \text{Range}(\text{Exp})$, it holds $\Pr[\text{Exp}(D, f) \in S] \leq e^\epsilon \cdot \Pr[\text{Exp}(D', f) \in S]$.

Definition 3.1 is motivated by the composition theorem, as our approach is aimed at a privately computed f . The resulting overall privacy guarantee for clustering and explanation is as follows. Let $M_{\text{clust}}(D, C)$ be a DP clustering mechanism, i.e., an algorithm that outputs a clustering function f while satisfying ϵ_{clust} -DP (e.g., DP k -means [56], where f is defined by the centers). By Proposition 2.8 (sequential composition), the sequential, adaptive composition $\text{Exp}(D, M_{\text{clust}}(D, C))$ satisfies $(\epsilon_{\text{exp}} + \epsilon_{\text{clust}})$ -DP. The sequential composition property for $\text{Exp}(\cdot, f)$ is applied with respect to its first argument for any function f within the range of M_{clust} .

A challenge with existing approaches is that previously proposed score functions for evaluating the quality of HBEs are highly sensitive, and hence cannot be used directly in the DP setting (see Section 4, and in particular Proposition 4.1). Moreover, these sensitive score functions are applied to pre-computed explanation candidates (Definition 2.3). In our setting, privately computing explanation candidates which are not selected for the output incurs a significant waste of privacy budget. Additionally, assessing candidate quality scores based on noisy histograms introduces more noise than necessary, as each bin is injected with independent noise, which accumulates and leads to inaccurate quality assessment. Instead, one would hope to assess *attribute* quality directly, and generate full explanations, as in Definition 2.3, only for selected attributes.

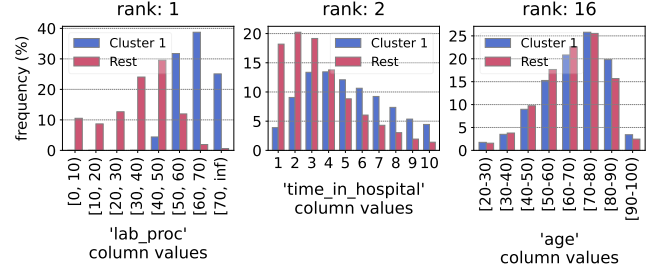


Figure 4: Ranked explanation candidates for Cluster 1 of the Diabetes dataset.

An *attribute combination* is a mapping $\mathcal{AC} : C \rightarrow \mathcal{A}$ that maps each cluster label to an attribute. Thus, our goal is to find a high-quality attribute combination such that the histograms of the corresponding attributes form a high-quality HBE¹. This leads to our first goal: deriving utility-wise equivalent but less sensitive *candidate attribute* quality functions. Then, our next goal is to develop a privacy-preserving algorithm that leverages the low-sensitivity score to produce high-quality explanations.

We can now summarize the challenges addressed in this work:

Problem 1 (Low Sensitivity Quality Functions). *Find a low-sensitivity, global quality function GIScore that maps a sensitive dataset D , a clustering function $f : \text{dom}(R) \rightarrow C$, and an attribute combination $\mathcal{AC}$ to a real number.*

Once we have a low-sensitivity score function, we can rank the explanations and return the highest scoring ones. Nevertheless, we still need a private mechanism to allow us to do so with low error.

Problem 2 (Select Top Explaining Attributes). *Given a sensitive dataset D , a clustering function $f : \text{dom}(R) \rightarrow C$, and a privacy budget ϵ , find a high-scoring attribute combination $\mathcal{AC}$ according to the global score function GIScore and output the corresponding global explanation while satisfying ϵ -DP.*

Example 3.2. Figure 4 gives an example of 3 explanation candidates for Cluster 1 of the Diabetes dataset. In this example, the top-ranked candidate is also selected to explain Cluster 1 in the final output, showcased in Example 1.1.

In the next sections, we describe our solution to Problem 1 and then describe our approach for obtaining high-scoring explanations (Definition 2.5) while satisfying ϵ -DP to solve Problem 2.

4 Low Sensitivity Quality Functions

In this section, we address Problem 1 by building upon prior work [7–9, 28, 66] and focusing on three prominent quality aspects of histogram-based explanations: Sufficiency [7, 8], interestingness [28, 54], and diversity [7, 66]. We prove that these measures are highly sensitive, making them unsuitable for a DP algorithm. These results then motivate us to devise low-sensitivity variants.

4.1 Interestingness

The interestingness of a histogram explanation is quantified as the distributional shift of the given attribute values between the cluster

¹While the global explanation uses one histogram per cluster, as in prior work [7], our framework can be extended to output multiple histograms per cluster. However, this comes at the cost of increased complexity, as further discussed in Appendix B.

and the full-dataset [7, 9, 28]. There are many ways to quantify distance between probability distributions.

For a dataset D and a cluster $D_c \subseteq D$, the total variation distance between the distributions of values in $\pi_A(D)$ and $\pi_A(D_c)$, and thus the *interestingness* of an attribute A with respect to the cluster D_c is defined as

$$\text{dist}_{TV}(\pi_A(D), \pi_A(D_c)) = \frac{1}{2} \sum_{a \in \text{dom}(A)} \left| \frac{\text{cnt}_{A=a}(D)}{|D|} - \frac{\text{cnt}_{A=a}(D_c)}{|D_c|} \right| \quad (1)$$

The global interestingness of an attribute combination $\mathcal{AC}$, is defined as the average of all single-cluster interestingness scores [7], which we denote by $\text{Int}(D, f, \mathcal{AC})$

In the context of differential privacy, *we cannot make any assumptions about the input dataset D or the clustering function*, and privacy must be guaranteed for any input, regardless of the cluster size. In particular, when a cluster is small, the removal of a single tuple from it can significantly change the cluster's internal distribution, which may lead to a large change in the distribution distance, as reflected in Equation (1). Given that sensitivity analysis must account for such cases, the sensitivity of this function is high.

PROPOSITION 4.1. *The sensitivity of dist_{TV} is at least $\frac{1}{2}$ and its range is $[0, 1]$.*

Intuitively, since this function outputs a number between 0 and 1, its sensitivity is relatively high. While the proof of Proposition 4.1 is provided in [1], we illustrate the issue in the following example.²

Example 4.2 (The Issue with Interestingness sensitivity). Suppose D is a dataset of size 100,000 with a binary attribute A . Suppose further that 95% of individuals in the dataset we have $A = 1$. Assume that the clustering is imbalanced, with a cluster D_c that contains only 1 tuple t , and $t[A] = 0$. In this case, from Equation (1), the interestingness score of attribute A for explaining cluster D_c is $\frac{1}{2}(|0.95 - 0| + |0.05 - 1|) = 0.95$. However, suppose a new tuple t' that satisfies $t'[A] = 1$ is added to cluster D_c . Now the interestingness in Equation (1) becomes $\frac{1}{2}(|95,001/100,001 - 0.5| + |5000/100,001 - 0.5|) \approx 0.45$. Note that while we only added a single tuple, it led to a change of $\approx 0.95 - 0.45 = 0.5$ in the interestingness function.

Low-Sensitivity Interestingness. We propose a new interestingness function that is inspired by the interestingness of [7], but has lower sensitivity.

Definition 4.3 (Low Sensitivity Interestingness). For a dataset D and a clustering function $f : \text{dom}(R) \rightarrow C$, define the interestingness of an attribute A for explaining the cluster $D_c = \{t \in D \mid f(t) = c\}$ as

$$\text{Int}_p(D, f, c, A) = \frac{1}{2} \sum_{a \in \text{dom}(A)} \left| \text{cnt}_{A=a}(D_c) - \frac{|D_c|}{|D|} \cdot \text{cnt}_{A=a}(D) \right|.$$

This new interestingness function gives us more leeway for adding the necessary noise to satisfy DP, while not distorting the results too much (see Appendix A.1 for proof).

PROPOSITION 4.4. *The sensitivity of $\text{Int}_p(D, f, c, A)$ is 1 and its range is $[0, |D_c|]$.*

²Previous work has also considered the Jensen-Shannon distance [37] as an interestingness measure. We show that it is highly sensitive as well, making it unsuitable for the privacy setting. The proposition and proof appear in Appendix A.1.

When D_c is not empty, $\text{Int}_p(D, f, c, A) = |D_c| \cdot \text{dist}_{TV}(\pi_A(D), \pi_A(D_c))$, where dist_{TV} is the total variation distance (Equation (1)). Therefore, when the cluster is fixed, the ranking of attributes by their interestingness scores is identical to their ranking by total variation distance deviation, which is the sensitive interestingness. However, intuitively, the effect of small cluster sizes on the sensitivity is mitigated by the multiplication by $|D_c|$, as now the sensitivity is 1, but the function takes values in $[0, |D_c|]$.

4.2 Sufficiency

We begin by revisiting the notion of sufficiency (faithfulness) from previous work, and then introduce our modified sufficiency function. The motivation behind this measure is to ensure that a given explanation is relevant only to its associated cluster, capturing the concept of sufficiency. Dasgupta et al. [8] introduced an abstract definition of sufficiency (faithfulness) of *classifier explanations*. Informally, the sufficiency of an explanation at $t \in D$ is defined as the fraction of tuples assigned the same prediction as t , out of all tuples for which the explanation of t “holds”. In other words, if t is assigned an explanation that holds for another tuple, then that tuple should have the same classification as t . Then, a global sufficiency score is obtained by averaging the sufficiency value at all tuples. However, their framework assumes the existence of a binary relation indicating whether an explanation holds for a given tuple, and extending this idea to HBEs is not straightforward. Copul et al. [7] proposed an adaptation of the formula from [8], replacing the classifier prediction with cluster assignment, and the binary relation with a probability value derived from the HBE. Specifically, they quantify the extent to which an HBE, defined by an attribute combination $\mathcal{AC}$, “holds for” a tuple t that belongs to a cluster D_c as the probability that uniformly random tuple sampled from D conditioned on having the same value in $\mathcal{AC}(c)$ as t , belongs to the same cluster as t . Unfortunately, this function, which we denote $\text{Suf}(D, f, \mathcal{AC})$, is too sensitive to be of use.³

PROPOSITION 4.5. *The sensitivity of $\text{Suf}(D, f, \mathcal{AC})$ is at least $\frac{1}{2}$ and its range is $[0, 1]$.*

Low Sensitivity Sufficiency. Next, we introduce an alternative formulation of the sufficiency measure for the following reasons. First, the original function is too sensitive for the DP setting. Second, the alternative formulation demonstrates that single-attribute sufficiency can, in fact, be measured for each individual cluster, with the global sufficiency being the average of these single-cluster sufficiency functions. This simplifies the sensitivity analysis, and implies that the globally most sufficient attribute combination can be obtained by independently selecting the most sufficient attribute for each cluster, a property leveraged in our algorithm. Third, it ensures that the range of each single-cluster sufficiency matches that of the interestingness measure, $[0, |D_c|]$ for a cluster D_c , and that the sensitivity of the modified sufficiency is also 1, making the two directly comparable.

³The high sensitivity of the sufficiency measure arises because we use the unbounded-DP variant, where the supremum in the definition of sensitivity is taken over all dataset sizes. This issue would not occur in the bounded-DP variant. However, there is still motivation to modify this function due to the other reasons outlined in this section.

Definition 4.6 (Low Sensitivity Sufficiency). For a dataset D and a clustering function $f : \text{dom}(R) \rightarrow C$, define the sufficiency of an attribute A for explaining the cluster D_c as

$$\text{Suf}_p(D, f, c, A) = \sum_{a \in \text{dom}_{D_c}(A)} \frac{\text{cnt}_{A=a}(D_c)^2}{\text{cnt}_{A=a}(D)}.$$

Note that we do not divide by zero, since the sum is only over values appearing at least once in D_c , and hence also in D .

Our modified sufficiency is consistent with the definition in previous work, as the equality in item (1) of Proposition 4.7 ensures that both measures induce the same ranking of attribute combinations when the dataset and clustering are fixed. We further prove a sensitivity bound of 1 for the single-cluster sufficiency function.

PROPOSITION 4.7. For an attribute combination $\mathcal{AC}$,

(1) The following equality holds:

$$|D| \cdot \text{Suf}(D, f, \mathcal{AC}) = \sum_{c \in C} \text{Suf}_p(D, f, c, \mathcal{AC}(c))$$

(2) The sensitivity of $\text{Suf}_p(D, f, c, A)$ is 1 and its range is $[0, |D_c|]$.

4.3 Diversity

The goal of the diversity measure is to quantify the overall distinctiveness among explanations. We provide an intuitive introduction of the diversity measure, denoted Div , outline its inadequacies in the context of DP, and discuss its relation to our new measure.⁴

Diversity is initially defined for a pair of single-cluster explanations and is subsequently generalized to global explanations by averaging all pairwise diversities. When two single-cluster explanations utilize different attributes, the diversity for that pair attains its maximum value of 1. If the explanations use the same attribute, the diversity is measured as the distance between the two distributions, quantifying the new knowledge gained from the additional explanation on that attribute. However, common metrics used in previous work, such as total-variation distance and Jensen-Shannon distance, have high sensitivity (Proposition 4.1, Footnote 2), which intuitively means they are unsuitable for the DP setting.

Inspired by the diversity measure of [7], we introduce the following low-sensitivity pairwise diversity function:

Definition 4.8 (Pair Diversity). For a dataset D and a clustering function f , the diversity score of a pair of attributes $A_c, A_{c'} \in \mathcal{A}$, where A_c (respectively $A_{c'}$) is a candidate attribute for explaining $D_c = \{t \mid f(t) = c\}$ (respectively $D_{c'}$), is

$$d(D, f, c, c', A_c, A_{c'}) = \min\{|D_c|, |D_{c'}|\} \times$$

$$\begin{cases} 1 & A_c \neq A_{c'} \\ \frac{1}{2} \sum_{a \in \text{dom}(A)} \left| \frac{\text{cnt}_{A=a}(D_c)}{\max\{|D_c|, 1\}} - \frac{\text{cnt}_{A=a}(D_{c'})}{\max\{|D_{c'}|, 1\}} \right| & A_c = A_{c'} \end{cases}$$

Note that when D_c and $D_{c'}$ are not empty and are explained by the same attribute A , we have

$$d(D, c, c', A, A) = \min\{|D_c|, |D_{c'}|\} \cdot \text{dist}_{TV}(\pi_A(D_c), \pi_A(D_{c'})).$$

Previous works on result diversification often define the overall diversity of a set of items using the average of pairwise diversities (e.g., [5, 61]). We adopt this approach in our definition of global diversity. We note that achieving low sensitivity requires that pairs

⁴The reader is referred to [7] for the original definition, and to [1] for our sensitivity analysis of this function.

from smaller clusters have a reduced impact on the global diversity function, as is evident in Definition 4.9. The sensitivity bound for this function is provided in Proposition 4.10, leveraging the fact that a convex combination of sensitivity-1 functions has a sensitivity bounded by 1 (see Lemma A.3).

Definition 4.9 (Global Diversity). For a dataset D and a clustering function $f : \text{dom}(R) \rightarrow C$, define the diversity score of an attribute combination $\mathcal{AC}$ as

$$\text{Div}_p(D, f, \mathcal{AC}) = \frac{1}{\binom{|C|}{2}} \sum_{\{c, c'\} \subseteq C} d(D, f, c, c', \mathcal{AC}(c), \mathcal{AC}(c'))$$

where the sum is over all distinct pairs of cluster labels.

PROPOSITION 4.10. The sensitivity of Div_p is bounded by 1. Moreover, its range is $[0, R_{\text{Div}}]$ where

$$R_{\text{Div}} = \frac{1}{\binom{|C|}{2}} \sum_{i=1}^{|C|} (|C| - i) |D_{c_i}|$$

is a weighted average of the cluster sizes, and $|D_{c_i}| \leq |D_{c_{i+1}}|$.

4.4 Combining All Quality Functions

We apply a common approach taken by multiple previous works on explainability (e.g., [7, 34, 39, 48, 53]). Specifically, we combine the different measures into a weighted sum, where the weights can be user-defined or adjusted based on specific requirements or preference. We first define the single-cluster score function, measuring the quality of an attribute A as an explaining attribute for a given cluster D_c .

Definition 4.11 (Single-Cluster Score). Let $\gamma = (\gamma_{\text{Int}}, \gamma_{\text{Suf}})$ be a pair of non-negative parameters that sum to 1. For a dataset D and a clustering function $f : \text{dom}(R) \rightarrow C$, define the quality score $A \in \mathcal{A}$ as a candidate attribute for a cluster $c \in C$, as

$$\text{Score}_\gamma(D, f, c, A) = \gamma_{\text{Int}} \cdot \text{Int}_p(D, f, c, A) + \gamma_{\text{Suf}} \cdot \text{Suf}_p(D, f, c, A).$$

Importantly, we can bound the sensitivity of the score function.

PROPOSITION 4.12. $\text{Score}_\gamma(D, f, c, A)$ has sensitivity bounded by 1 and its range is $[0, |D_c|]$.

The global score combines individual measures (as a weighted average) to assess the quality. The influence of a cluster on the global score, through its single-cluster interestingness or pairwise diversity, is proportional to its size. Smaller clusters contribute less than larger ones for the same distribution distance, addressing the high sensitivity of the original global score. As shown in Proposition 4.1, removing a single point from a small cluster can significantly alter its column distribution, leading to a large score change. In Appendix A.4, we prove the sensitivity bound Proposition 4.14.

Definition 4.13 (Global Score). Let $\lambda = (\lambda_{\text{Div}}, \lambda_{\text{Int}}, \lambda_{\text{Suf}})$ be non-negative parameters that sum to 1. For a dataset D and a clustering function f , define the overall quality score of an attribute combination $\mathcal{AC}$ as

$$\text{GLScore}_\lambda(D, f, \mathcal{AC}) = \lambda_{\text{Int}} \cdot \text{Int}_p(D, f, \mathcal{AC}) + \lambda_{\text{Suf}} \cdot \text{Suf}_p(D, f, \mathcal{AC}) + \lambda_{\text{Div}} \cdot \text{Div}_p(D, f, \mathcal{AC})$$

where we extend $\text{Int}_p(D, f, \mathcal{AC}) = \frac{1}{|C|} \sum_{c \in C} \text{Int}_p(D, f, c, \mathcal{AC}(c))$ and $\text{Suf}_p(D, f, \mathcal{AC}) = \frac{1}{|C|} \sum_{c \in C} \text{Suf}_p(D, f, c, \mathcal{AC}(c))$.

PROPOSITION 4.14. GScore_λ has sensitivity bounded by 1. Moreover, its range is $[0, R_{\text{GScore}_\lambda}]$ where

$$R_{\text{GScore}_\lambda} = (\lambda_{\text{Int}} + \lambda_{\text{Suf}}) \cdot \frac{1}{|C|} \sum_{c \in C} |D_c| + \lambda_{\text{Div}} \cdot R_{\text{Div}}$$

is a weighted average of the cluster sizes, and R_{Div} is defined as in Proposition 4.10.

Selecting the Weight Parameters. Previous work on HBEs in the non-private setting [7] has shown through user studies and quantitative evaluations that equal weight distribution $\lambda_{\text{Int}} = \lambda_{\text{Suf}} = \lambda_{\text{Div}} = 1/3$ produces high quality and informative explanations. We adopt this default parameter setting in our framework. However, weights can be adjusted based on preference. Our experiments show that DPclustX maintains high explanation quality compared to the non-private baseline across different weight distributions.

5 The DPclustX Framework

In this section, we address Problem 2, and introduce our algorithms for computing explanations (depicted in Figure 3). Since the space of all possible attribute combinations is generally too large to analyze, we propose a novel DP-tailored manner of pruning the search space. Following previous work [7], we construct a small set of high-quality candidate *attributes* for each cluster (Stage-1), which will later serve as the candidate pool for a *global* explanation of all clusters (Stage-2). To optimize our privacy budget usage, we avoid generating noisy histograms at Stage-1.

5.1 Stage-1: Construct Private Candidate Sets

We now present our candidate set construction algorithm (Figure 3).

In DPclustX, to satisfy DP, we privately select top- k candidate *attributes* for each cluster based on our single-cluster score function. Since doing so by iteratively applying the exponential mechanism k times can be computationally expensive, we adopt the *One-shot Top- k mechanism* [13] to privately select top- k . This mechanism adds independent Gumbel noise⁵ to each of the true scores with scale $\sigma = 2\Delta_{\text{Score}} \cdot k/\epsilon$, reorders all the candidates in a descending order by their noisy scores, and outputs the first k candidates. It satisfies ϵ -DP since it is identical to iteratively applying k exponential mechanisms [13], where each satisfies ϵ/k -DP. Therefore, by sequential composition (Proposition 2.8) it satisfies overall ϵ -DP.

The pseudo code in Algorithm 1 provides the candidate selection procedure. we first set the privacy budget $\epsilon_{\text{Topk}} = \epsilon_{\text{CandList}}/|C|$ allocated for each Top- k selection, and in Line 2 we set the noise scale parameter $\sigma = 2k/\epsilon_{\text{Topk}}$. Then, in Lines 3-9, we run the Top- k selection procedure described above for each cluster $c \in C$, obtaining a set S_c of Top- k attributes for each cluster. Specifically, in Line 5 we compute the noisy single-cluster score (Definition 4.11) for each cluster and attribute. In Line 7, we sort that attributes based on the noisy scores for each cluster. In Line 9, we define the set S_c for each cluster $c \in C$. The output returned in Line 11 consists of the sets S_c for every $c \in C$.

Note that $1/|C|$ -fraction of the total privacy budget $\epsilon_{\text{CandList}}$ is allocated for selecting top candidate attributes for each cluster. While one might have hoped to use parallel composition, this is typically not possible. The quality score of an attribute for a given

cluster depends on the entire dataset, not just the cluster itself. A high-scored attribute exhibits large distributional deviation between the cluster and the full dataset, requiring consideration of tuples outside the cluster.

Algorithm 1 Select-Candidates: Generate all single-cluster top- k candidate attributes

Input: Dataset D , clustering function $f : \text{dom}(R) \rightarrow C$, hyperparameters $\gamma = (\gamma_{\text{Int}}, \gamma_{\text{Suf}})$ Attribute set $\mathcal{A}$, privacy parameter $\epsilon_{\text{CandList}}$, cardinality k .

Output: Sets $S_{c_1}, \dots, S_{c_{|C|}}$ where S_{c_i} contains noisy Top- k candidate explaining attributes for D_{c_i} .

```

1:  $\epsilon_{\text{Topk}} \leftarrow \epsilon_{\text{CandList}}/|C|$ 
2: Set  $\sigma \leftarrow 2k/\epsilon_{\text{Topk}}$ 
3: for  $c \in C$  do
4:   for  $A \in \mathcal{A}$  do
5:      $s_A \leftarrow \text{Score}_\gamma(D, f, c, A) + \text{Gumbel}(\sigma)$ 
6:   end for
7:   Sort  $\{s_A \mid A \in \mathcal{A}\}$  in descending order.
8:   Let  $A_1, \dots, A_k$  be the attributes corresponding to the top- $k$ 
      noisy scores.
9:   Set  $S_c \leftarrow \{A_1, \dots, A_k\}$ 
10: end for
11: return Candidate sets  $S_{c_1}, \dots, S_{c_{|C|}}$ .
```

The privacy and utility guarantees of Algorithm 1 are given by the following proposition:

PROPOSITION 5.1. Given a clustering function f , a set of attributes $\mathcal{A}$, a privacy parameter $\epsilon_{\text{CandList}}$, non-negative hyperparameters $\gamma_{\text{Int}}, \gamma_{\text{Suf}}$ that sum to 1, and a size parameter k , the following holds:

- (1) Algorithm 1 satisfies $\epsilon_{\text{CandList}}$ -DP.
- (2) For each $c \in C$, denote by $\text{OPT}_c^{(\ell)}$ the ℓ -th highest (true) score, and by $A_c^{(\ell)}$ the ℓ -th explaining attribute selected by Algorithm 1 to S_c . For all c and for all $\ell = 1, 2, \dots, k$, we have

$$\Pr \left[\text{Score}(A_c^{(\ell)}) \leq \text{OPT}_c^{(\ell)} - \frac{2|C| \cdot k}{\epsilon_{\text{CandList}}} (\ln |\mathcal{A}| + \epsilon) \right] \leq e^{-\epsilon}.$$

The proof of item 1 follows from the sequential composition theorem (Proposition 2.8) and the analysis of the one-shot Top- k mechanism [12, 13]. The proof of item 2 is based on the utility proposition of exponential mechanism (Theorem 3.11 in [17]). Although the utility bound may appear pessimistic under certain parameter regimes, our experiments in Section 6 show that our mechanism achieves optimal or near-optimal explanations even for small privacy budgets.

Example 5.2. Reconsider Example 1.1. Algorithm 1 outputs the set $S_1 = \{\text{lab_proc}, \text{time_in_hospital}, \text{num_medications}\}$, which indeed comprises the top-3 attributes for Cluster 1 obtained from the ranked list partly presented in Figure 4. Algorithm 1 outputs one such set for each cluster.

5.2 Stage-2: A Private Global Explanation

We now present a privacy-preserving mechanism that, given a dataset D , a clustering function f , outputs a histogram-based explanation for the given clustering, selects the explaining attribute

⁵The CDF of the Gumbel distribution $\text{Gumbel}(\sigma)$ is $F(z) = \exp(-\exp(-z/\sigma))$.

for each cluster D_c from its corresponding set S_c (Section 5.1), as depicted in Figure 3.

As in Stage-1, pre-computing noisy histograms and selecting the best candidate combination afterward incurs a waste of privacy budget, and evaluating the global score based on noisy histograms introduces more noise than necessary due to the injection of noise into *each count* of the private histograms. To address these issues, we propose the following approach.

We begin by computing the single-cluster candidate sets S_c using Algorithm 1 with privacy budget $\epsilon_{CandList}$. Then, we apply the exponential mechanism to select the noisy-best *attribute combination* from a global candidate set, drawn from each cluster's candidate set, and using our global score function (Definition 4.13). Subsequently we generate the noisy histograms *only for the selected attribute combination* $\mathcal{AC}$, employing a black-box mechanism for histogram generation while leveraging the parallel composition property for efficient privacy budget allocation.

The pseudo code describing our approach can be found in Algorithm 2. We first set the marginal weights of interestingness and sufficiency of the single-cluster score function. In Line 3, we invoke Algorithm 1, obtaining the single-cluster candidate sets S_c for all $c \in C$. In Line 6, we run the exponential mechanism with the global score function, GScore_λ (Definition 4.13), a privacy parameter $\epsilon_{TopComb}$, and the set of possible candidate attribute combinations, mapping each cluster label c to an attribute from the candidate set S_c . In Line 8, we compute the set of attributes appearing at least once in the combination. In Line 9, we set $\epsilon_{hist,all}$ as $\epsilon_{Hist}/(2|\mathcal{A}'|)$ and $\epsilon_{hist,cluster}$ as $\epsilon_{Hist}/2$, which are the privacy budgets for the full-dataset and single-cluster histogram generation, respectively. In Lines 10–12, we invoke a black-box histogram generating mechanism to compute all full-dataset histograms. In Line 16 we compute all cluster histograms. In Line 17 we compute the out-of-cluster histogram by subtracting the cluster noisy histogram from the full-dataset noisy histogram and replacing any negative entries with 0. In Line 18, we set each single-cluster explanation e_c using the computed histograms. In Line 20, we output the global explanation $\{e_c \mid c \in C\}$.

The following theorem states the privacy guarantee of DPCLustX.

THEOREM 5.3. *Given a clustering function $f : \text{dom}(R) \rightarrow C$, number of candidates k , privacy parameters $\epsilon_{CandList}, \epsilon_{TopComb}, \epsilon_{Hist}$, Algorithm 2 is $(\epsilon_{CandList} + \epsilon_{TopComb} + \epsilon_{Hist})$ -DP.*

PROOF SKETCH. The execution of Algorithm 1 satisfies $\epsilon_{CandList}$ -DP by Proposition 5.1. Since the function GScore_λ has sensitivity 1 (Proposition 4.14), the execution of the exponential mechanism in Line 6 guarantees $\epsilon_{TopComb}$ -DP, applying Theorem 2.11. We assume that each execution of $\mathcal{M}_{hist}(\cdot, \epsilon)$ satisfies ϵ -DP. Therefore, by sequential composition, the computation of $\tilde{h}_A$ for all $A \in \mathcal{A}'$ satisfies overall $\epsilon_{Hist}/2$ -DP. Since the clusters are disjoint, the computation of all noisy histograms $\tilde{h}^c$ satisfies overall $\epsilon_{Hist}/2$ -DP by parallel composition. Note that the computation of each $\tilde{h}^{-c}$ in Line 17 is post-processing, and therefore incurs no additional privacy loss. Overall, by sequential composition, the entire algorithm guarantees $(\epsilon_{CandList} + \epsilon_{TopComb} + \epsilon_{Hist})$ -DP. $\square$

Algorithm 2 DPCLustX Algorithm - Generate Global Explanation

Input: Dataset D , clustering function f , number of candidates k , privacy parameters $\epsilon_{CandList}, \epsilon_{TopComb}, \epsilon_{Hist}$, hyperparameters $\lambda_{Div}, \lambda_{Int}, \lambda_{Suf}$

Output: Global explanation $\{e_c \mid c \in C\}$ (Definition 2.5)

```

1: Let  $\gamma_{Suf} \leftarrow \lambda_{Suf}/(\lambda_{Suf} + \lambda_{Int})$  and  $\gamma_{Int} \leftarrow \lambda_{Int}/(\lambda_{Suf} + \lambda_{Int})$ 
2: // Invoke Algorithm 1
3: Let  $S_{c_1}, \dots, S_{c_{|C|}} \leftarrow \text{Select-Candidates}(D, f, \gamma, \mathcal{A}, \epsilon_{CandList}, k)$ 
4: // Select attribute combination
5: Let  $\mathcal{R} \leftarrow \{\mathcal{AC} \mid \forall c. \mathcal{AC}(c) \in S_c\}$  be the set of candidate combinations, mapping each  $c \in C$  to an attribute in  $S_c$ .
6:  $\mathcal{AC} \leftarrow \mathcal{M}_E(D, \text{GScore}_\lambda, \Delta_{\text{GScore}}, \mathcal{R}, \epsilon_{TopComb})$ 
7: // Generate full-dataset histograms
8: Let  $\mathcal{A}' \leftarrow \{\mathcal{AC}(c) \mid c \in C\}$  be the set of attributes appearing in at least once in the combination.
9:  $\epsilon_{hist,all} \leftarrow \epsilon_{Hist}/(2|\mathcal{A}'|)$ ,  $\epsilon_{hist,cluster} \leftarrow \epsilon_{Hist}/2$ 
10: for every  $A \in \mathcal{A}'$  do
11:    $\tilde{h}_A \leftarrow \mathcal{M}_{hist}(\pi_A(D), \epsilon_{hist,all})$ 
12: end for
13: // Compute single-cluster explanations
14: for every  $c \in C$  do
15:   Let  $A_c \leftarrow \mathcal{AC}(c)$  be the attribute selected for  $D_c$ .
16:    $\tilde{h}^c \leftarrow \mathcal{M}_{hist}(\pi_{A_c}(D_c), \epsilon_{hist,cluster})$ 
17:    $\tilde{h}^{-c} \leftarrow \max(\tilde{h}_{A_c} - \tilde{h}^c, 0)$ 
18:    $e_c \leftarrow (c, A_c, \tilde{h}^{-c}, \tilde{h}^c)$ 
19: end for
20: return global explanation  $\{e_c \mid c \in C\}$ .
```

Example 5.4. Consider the setting in Example 1.1, and suppose the input contains 3 clusters of the Diabetes dataset. First, a candidate set is obtained for each cluster, with the set for Cluster 1 shown in Example 5.2. In Line 6, the selected attribute combination is: Cluster 1: lab_proc, Cluster 2: lab_proc, and Cluster 3: discharge_disp. DP noisy histograms are generated for these attributes. Figure 2a depicts a part of the output generated by Algorithm 2, showcasing the explanation for Cluster 1. The full output of Algorithm 2 provides such an explanation for each cluster.

6 Experiments

In this section, we evaluate the quality and efficiency of the explanations generated by DPCLustX with the following questions:

- (1) With respect to the quality measures for HBes, how does DPCLustX perform compared to the non private method and a naive approach which computes all histograms in advance?
- (2) How close is the attribute combination selected by DPCLustX to that of the non-private baseline?
- (3) How efficient is DPCLustX in computing the explanations?

Summary of our findings. With a total privacy budget of $\epsilon = 0.1$, DPCLustX selects attributes of comparable quality to those chosen by the non-private baseline in all datasets and clustering methods. Moreover, at $\epsilon = 1$, DPCLustX consistently selects the same attributes as the non-private baseline across all runs and clustering methods for the Diabetes dataset. The execution time of DPCLustX for generating explanations averages under 6.6 seconds across all datasets and clustering methods with at most 9 clusters.

6.1 Experimental Settings

We next present our settings for the experiments. We have implemented DPclustX in Python 3.9.19 using the Pandas and NumPy libraries. All experiments were run on an Intel Xeon CPU-based server with 24 cores and 96 GB of RAM. We use the Geometric mechanism [24] for DP histogram generation, implemented by DiffPrivLib [29]. The source code is available in [1].

Default parameters. Unless mentioned otherwise, the following default parameters are used. We set $\epsilon_{CandList} = \epsilon_{TopComb} = \epsilon_{Hist} = 0.1$. Thus, the combined privacy budget for attribute selection is $\epsilon = 0.2$, which we vary from 0.001 to 1 when evaluating its effect on the selected attributes. The number of candidate attributes per cluster selected at Stage-1 (denoted by k in Section 5.1) is set to 3, a choice supported by ablation studies presented in Figure 7, where it is varied from 1 to 5. Following the discussion in Section 4.4, we set the default values $\lambda_{Int} = \lambda_{Suf} = \lambda_{Div} = 1/3$. We also evaluate alternative weight distributions, where we set one weight to zero and each of the remaining two to $1/2$.

Datasets. We evaluate DPclustX on the following two datasets:

- **US Census Data** [45]: 2,458,285 tuples and 68 attributes. This dataset contains a one percent sample of the Public Use Microdata Samples (PUMS) person records drawn from the 1990 census.
- **Diabetes** [6]: 101,766 tuples and 47 attributes. This dataset comprises ten years (1999–2008) of clinical care data. Each tuple represents a hospital record of a diabetic patient who underwent laboratory tests, received medications, and had a hospital stay of up to 14 days. The objective is to predict whether a patient will be readmitted within 30 days of discharge. Numerical and large-domain categorical attributes are binned to ensure domain sizes smaller than 40. Details can be found in [1].

Clustering methods. We assessed various clustering algorithms to demonstrate the effectiveness and versatility of DPclustX across a wide range of scenarios. This evaluation highlights the robustness of our method in providing meaningful explanations for different clustering tasks, including: (i) k -means, (ii) DP- k -means [56] with $\epsilon = 1$, implemented by DiffPrivLib [29], (iii) k -modes, (iv) Agglomerative Clustering, (v) Gaussian Mixture. For each method, we evaluate our framework on 3, 5, 7 and 9 clusters with each method. For the purpose of clustering, categorical attributes we transformed into equivalent numerical data by mapping each domain value to a unique integer. Due to its scalability limitations, Agglomerative clustering is not applied to the Census dataset.

To cover a broad spectrum of scenarios, we consider both private and non-private clustering methods, aligning with the approach taken in evaluations of differentially private black-box classifier explanations [26, 46, 51]. We note that in real-world deployment, to protect data privacy, our framework must be provided with either a privately computed or a data-independent clustering function.

Baselines. We compare the performance of DPclustX with the following approaches for generating explanations:

- **TabEE (non-private):** The non-private algorithm in [7] for finding a high-scoring attribute combination. The algorithm selects the top attribute combination from a pre-constructed candidate pool based on the original, sensitive definition of the quality functions, as we discussed in Section 4.

- **DP-Naive:** In Section 5 we discussed a naive approach for computing HBEs under DP. We implement this simple DP baseline as follows. Given a privacy budget ϵ , we compute each of the full-dataset histograms using a budget $\epsilon/(2|\mathcal{A}|)$ for each attribute. We compute the histogram of each cluster for each attribute using a budget of $\epsilon/(2|\mathcal{A}|)$ per cluster. Then, as a post-processing step, we run the TabEE-based algorithm on the noisy histograms. By the composition and post-processing theorems, the entire algorithm satisfies ϵ -DP.

Evaluation measures. While in this work we introduce low-sensitivity variants of *interestingness*, *sufficiency* and *diversity*, which are incorporated into our algorithm, the *original*, sensitive variants of these functions can still be used for evaluating the quality of the selected attribute combination. Hence, we let $Quality = \lambda_{Int} \cdot Int + \lambda_{Suf} \cdot Suf + \lambda_{Div} \cdot Div$ be the global score function from [7], using the sensitive quality functions Int, Suf, Div (Section 4).

We also evaluate the similarity between the attribute combination selected by non-private baseline, denoted $\mathcal{AC}^*$, and that of DPclustX or the DP-Naive baseline. To this end, we adapt the mean absolute error (MAE) [62] to our discrete setting. The MAE score for a combination $\mathcal{AC}$ is given by $MAE(E) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \mathbf{1}_{\{\mathcal{AC}(c) \neq \mathcal{AC}^*(c)\}}$.

6.2 Quality Analysis

We detail our experimental analysis for the different configurations of our framework. All results are averaged over 10 runs.

Selected attributes quality score. In this experiment (Figure 5), we aim to demonstrate that our algorithm selects high-quality attributes even under tight privacy budgets, achieving optimal or near-optimal results compared to the non-private baseline. We evaluate the global quality score of the selected attribute combination with default parameters for different privacy budgets ϵ , where $\epsilon_{CandList} = \epsilon_{TopComb} = \epsilon/2$. Note that this experiment examines the attribute choice, hence histogram generation is not needed. Figure 5 shows the trend of $Quality$ values for 5 clusters as ϵ varies for DPclustX (green), TabEE (dotted blue), and DP-Naive (dashed gray). Additional results for other cluster numbers are provided in Appendix C, showing similar trends.

We find that increasing the total privacy budget ϵ improves the quality scores in all cases. For the Diabetes dataset, at $\epsilon = 0.1$, DPclustX scores are 0.66% lower than TabEE on average, while DP-Naive scores are 20.22% lower. At $\epsilon = 1$, DPclustX matches TabEE across all methods. For the Census dataset, at $\epsilon = 10^{-2.5}$, DPclustX scores are 1.6% lower on average, while DP-Naive scores are 9.8% lower. At $\epsilon = 0.1$, the difference for DPclustX is only 0.003%.

Selected attributes error. This experiment (Figure 6) examines the extent to which DPclustX and the non-private baseline select similar attribute combinations, given their close quality scores. We compute the MAE for the attribute combinations selected by DPclustX with varying privacy budgets ϵ , where $\epsilon_{CandList} = \epsilon_{TopComb} = \epsilon/2$. Figure 6 shows the MAE trend for 5 clusters. Results indicate that DPclustX closely matches the non-private baseline across all methods, even for small privacy budgets. For $\epsilon = 1$, DPclustX selects the same combination as TabEE for all methods. For $\epsilon = 0.1$, MAE is below 0.04 for most methods, except k -modes (below 0.46). For

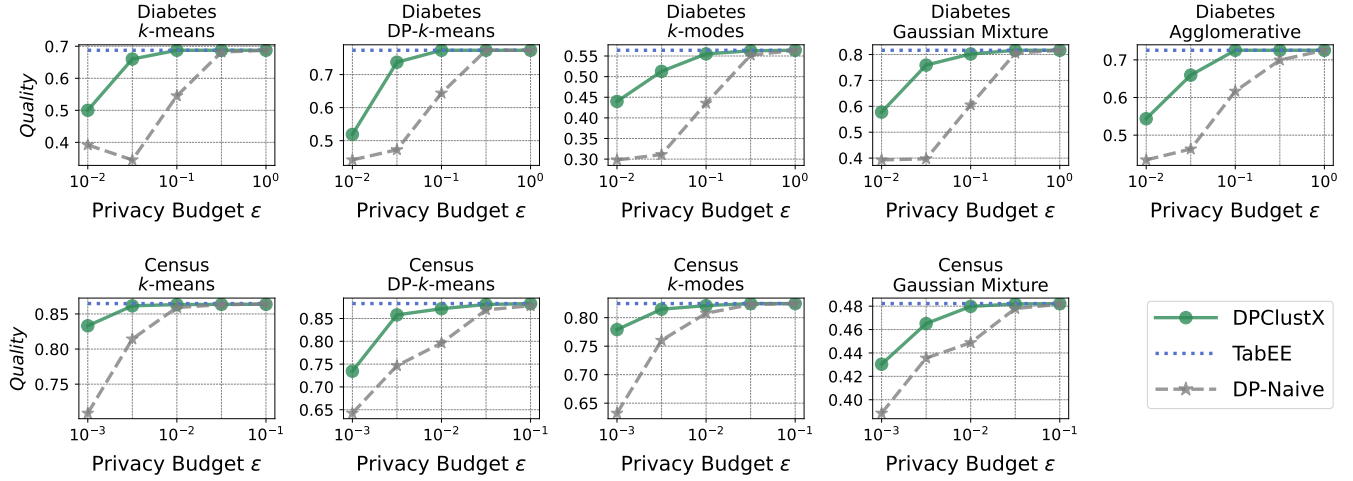


Figure 5: *Quality* values of the selected attribute combination with 5 clusters, as the total privacy budget ϵ varies.

$\epsilon \geq 0.1$ on the Diabetes dataset, MAE values are below 0.25 for k -means, 0.2 for k -modes, 0.04 for DP- k -means, and 0.08 for Gaussian Mixture. Notably, k -means does not approach 0 even at $\epsilon = 1$, as the clustering can be explained by correlated attributes. Similar case is discussed in Section 6.4. DP-ClustX also consistently outperforms DP-Naive in this measure as well.

Quality for different candidate set sizes. In this experiment (Figure 7), we evaluate the score of the selected attribute combination with varying candidate set sizes from Stage-1 of our algorithm. Figure 7 shows the *Quality* trend for 5 clusters. We find that increasing the candidate set size generally has either a positive or no effect on the score. For the Diabetes dataset and all methods except k -modes, increasing the candidate size from 1 to 5 had no effect, with the same attributes selected. For k -modes, the score increases by 8% between 1 and 3 candidates and stabilizes. For the Census dataset, a positive trend is observed for all methods between 1 and 3, peaking at 3 and stabilizing. Gaussian mixture shows a 40% score improvement between 1 and 2 candidates, with the same attributes selected for 3. Unlike the non-private setting, where increasing the candidate size improves quality, in the DP setting, quality may degrade with larger sets, particularly at low privacy budgets, when smaller sets already include highly diverse candidates. We set the default size to 3, as it performs best in most cases.

Quality for different choices of weights. We measure the quality score of the selected attribute combination for different weight parameters λ_{Int} , λ_{Suf} , and λ_{Div} , setting one to zero and the others to 1/2. The results show that for 5 clusters and the Diabetes dataset, scores are lower by 0.06% on average, with a maximum difference of 0.4% for Agglomerative clustering when $\lambda_{\text{Int}} = 0$. For the Census dataset, scores are lower by 0.13% on average, with a maximum difference of 1.08% for DP- k -means when $\lambda_{\text{Int}} = 0$. These results, detailed in [1], show that DP-ClustX selects an attribute combination with scores comparable to the non-private baseline, offering the same flexibility in parameter selection.

6.3 Performance Analysis

We measure the execution time of our algorithm with various parameter settings for both datasets. We focus on k -means and Gaussian Mixture clustering methods, as other methods were unable to scale to a large number of clusters or returned mostly empty clusters when the number of clusters exceeded 9. The default settings for these experiments are 9 clusters, 3 candidate attributes per cluster, and the entire dataset with all attributes used. Each experiment varies one of these parameters and is averaged over 10 runs.

Number of clusters. Figure 8a illustrates the relationship between the execution time (in seconds, log scale) and the number of clusters for which an explanation is generated. We observe that the running time increases exponentially with the number of clusters, yet remains reasonably low across both methods when the number of clusters is at most 11. For the larger Census dataset, the runtimes for up to 9 clusters across both methods are less than 6.6 seconds. For the dataset Diabetes, all runtimes are below 3 seconds up to 9 clusters, exhibiting a similar trend. For 15 clusters, the runtimes are up to 1275 seconds for Diabetes, and 1314 seconds for Census.

Number of candidate attributes per cluster. Figure 8b illustrates the relationship between the execution time (in seconds, log scale) and the size of the candidate set constructed by Algorithm 1 for each cluster. The results indicate a significant execution time increase with larger sizes, highlighting the benefit of selecting a smaller size. For the Diabetes dataset, execution times remain below 3 seconds with a maximum of 3 candidates. For the Census dataset, execution times stay under 7 seconds with at most 3 candidates. However, with 5 candidates, execution times rise up to 80 seconds for the Census dataset and 75 seconds for the Diabetes dataset.

Number of attributes. In this experiment (Figure 8c), we randomly sample a subset of the attributes for each dataset, and generate an explanation using only the sampled attributes. Figure 8c depicts the relationship between running time (in seconds) and the percentage of attributes selected from the dataset. The results indicate a linear increase in running time, implying that while the number of attributes has an effect, it is relatively small. For instance, with a 50%

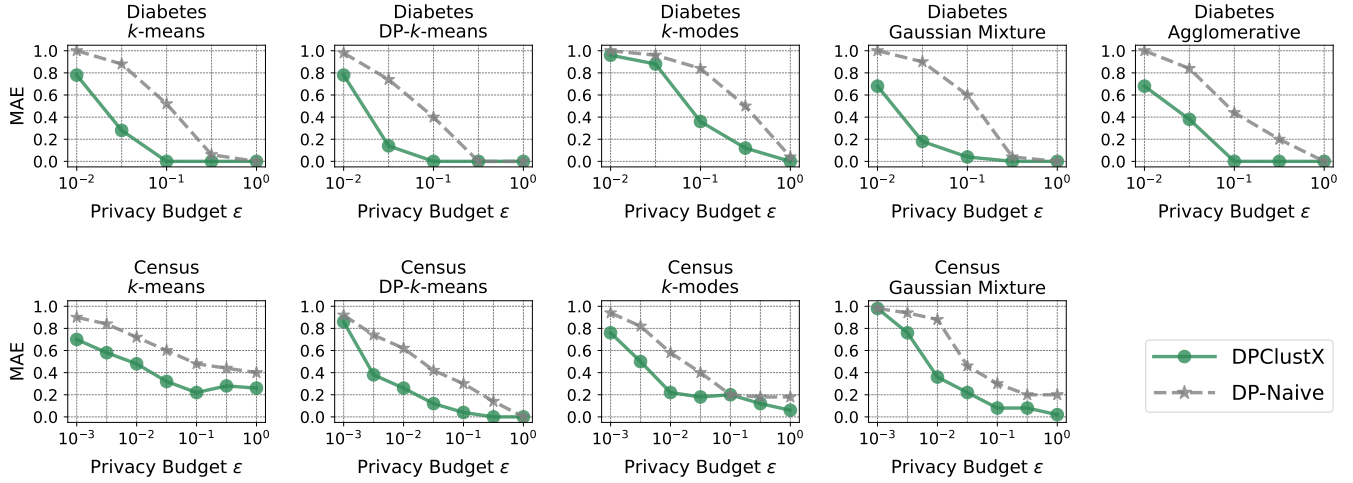


Figure 6: Mean Absolute Error (MAE) values of the selected attributes with 5 clusters compared to the non-private TabEE baseline [7], as the total privacy budget ϵ varies.

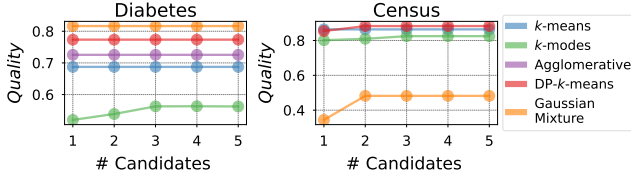


Figure 7: Explanation Quality values for DP-ClustX with a varying number of candidate attributes per cluster selected at Stage-1 (Section 5.1).

sample, the execution times are at most 4.6 seconds for the Census dataset and 2 seconds for Diabetes, while with a 100% sample (full dataset), the execution times increase to a maximum of 6.3 seconds for Census and 2.9 seconds for Diabetes.

Dataset size. In this experiment (Figure 8d), we randomly sample a portion of the tuples for each dataset, and generate an explanation using the sampled data. Figure 8d shows the relationship between execution time (in seconds) and the percentage of tuples sampled from each dataset. The results indicate a linear increase in running time, suggesting that while an increase in the number of tuples leads to longer execution times, the effect remains relatively small. With a 50% sample, the execution times are at most 4.8 seconds for the Census dataset and 2.6 seconds for Diabetes. With a 100% sample (full dataset), the execution times are at most 6.3 seconds for Census and 2.9 seconds for Diabetes.

6.4 Case Study

We now present a case study over the Census dataset with default parameters. The dataset is clustered into 3 clusters using *k*-means.

DP-ClustX selects the attribute combination [iLabor, iWork89, dHours]. The attribute iLabor represents employment status, iWork89 indicates whether the individual worked in 1989, and dHours denotes the number of hours the individual worked in the previous week. The final output, including the noisy histograms, is shown in Figure 9a. Figure 9b shows the non-private explanation generated by the TabEE baseline. In this case, the selected attributes

are [iLabor, iYearwrk, iMeans]. The attribute iYearwrk specifies the last year the individual worked, and iMeans specifies the means of transportation to work. Since the two explanations agree on 1 out of 3 attributes, DP-ClustX achieves $MAE = 2/3$. However, the *Quality* of the non-private baseline is only 0.04% higher, and the explanation generated by DP-ClustX conveys similar insights.

Both explanations indicate that Cluster 1 predominantly consists of adults who are currently working. For Cluster 2, both explanations suggest that it mostly represents individuals under age 16, for whom data is unavailable. In this group, the attributes iWork89 and iYearwrk are correlated. For Cluster 3, the DP-ClustX explanation highlights a high representation of individuals who worked more than 0 hours last week, implying they are employed. The non-private explanation also indicates that this cluster predominantly consists of working individuals, as they all have means of transportation to work. Thus, a similar conclusion is reached, though different attributes are used. Note that the attributes dHours and iMeans are correlated among non-working individuals.

7 Related Work

We next survey related work in relevant fields.

Differentially private explanations. Several works have explored integrating DP with ML models and query result explanation methods to address the dual challenge of transparency and privacy. The work of [51] introduced differentially private mechanisms for model explanations, providing interpretable insights into model behavior while satisfying DP. Their approach adapts traditional explanation methods such as feature importance scores and local interpretable model-agnostic explanations (LIME) to operate under DP constraints, thereby ensuring that the explanations themselves do not leak sensitive information. Additionally, significant contributions have been made in creating a framework for generating differentially private Shapley values, enabling the interpretation of model predictions with strong privacy guarantees [32]. Further approaches include a DP SVM mechanism for robust counterfactual explanations [46] and DP Locally Linear Maps, which provide both

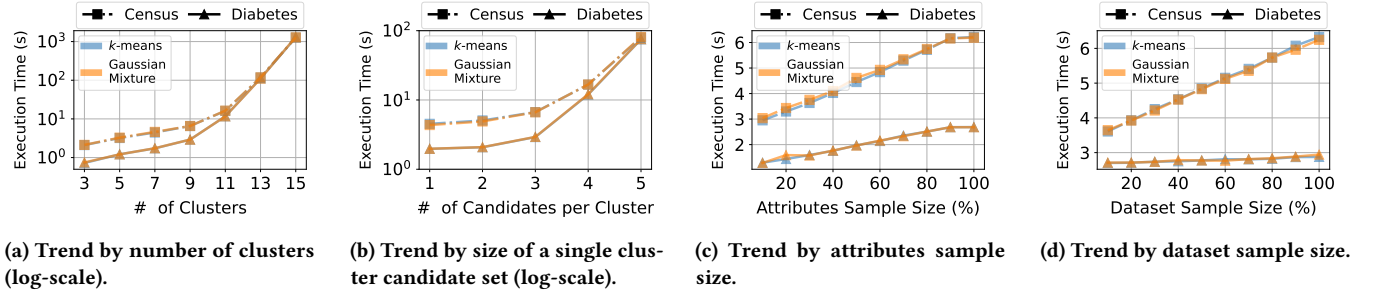


Figure 8: DPCLustX execution time experiments for the two datasets as functions of different parameters.

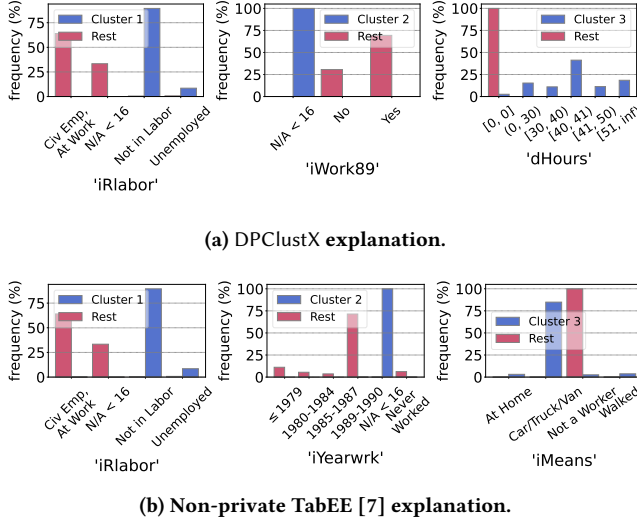


Figure 9: Explanations for the US Census dataset case study. local and global model explanations while enabling a favorable privacy-accuracy trade-off by efficiently managing the number of parameters [26]. See [49] for a survey on privacy-preserving model explanations. In the context of query result explanation, DPXPlain [59] is a framework that generates differentially private explanations for aggregate queries based on the notion of intervention. These efforts represent crucial steps towards creating trustworthy ML and data analysis systems that offer both transparency and robust privacy guarantees, thus addressing the critical need for ethical AI deployment in sensitive applications.

Clustering explanations. In the non-private setting, clustering explanations are well-studied. For instance, [19, 21, 40, 41, 47] study tree-based interpretable clustering algorithms specifically for k -means or k -medians. Tutay and Somech [60] provides black-box clustering explanations using filter predicates. Copul et al. [7] use histogram-based explanations for clusters in tabular datasets, induced by clusters formed in the tabular-embedding space. However, *differentially-private* explanation algorithms for clustering have not been explored beyond our work.

Private data summarization and exploration tools. In the realm of databases, privacy-preserving techniques have been a focal point of research, particularly with the advent of differential privacy (DP). In recent years, DP has been increasingly adopted to practical systems for interactive data analysis, such as PINQ [44], PrivateSQL

[33], FLEX [31], and Chorus [30]. Another significant contribution is the DPCube framework by Xiao et al., which enables efficient and private OLAP data cube construction, by differentially private histogram release through multidimensional partitioning, thereby enhancing the usability of summarized data under privacy constraints [64]. The RONA algorithm by McKenna et al. improves the accuracy of DP synthetic data generation through an iterative refinement process, supporting private data release [42]. Additionally, Zhang et al. introduced PrivBayes, a method for generating differentially private synthetic data using Bayesian networks to capture correlations in the data while preserving privacy [67]. In the context of data exploration, APEX [22] is a system that allows data analysts to pose adaptively chosen queries along with required accuracy bounds, identifying algorithms with the least privacy loss to answer these queries accurately under DP.

These advancements collectively highlight the progress and ongoing challenges in developing privacy-preserving database systems that balance data utility with stringent privacy requirements.

8 Conclusion and Future Work

We proposed DPCLustX, a framework for generating global, histogram-based explanations for black-box clustering results while preserving DP. These explanations consist of histograms on carefully selected attributes for each cluster and the remaining dataset, highlighting significant distributional shifts alongside additional high-quality characteristics. We proposed low sensitivity adaptations of these quality scores, provided their sensitivity analysis, and devised a privacy preserving mechanism for generating explanations. We demonstrated through extensive experiments that the explanations generated by DPCLustX are comparable to the non-privately generated explanations even under tight privacy budgets.

There are several interesting future directions. First, the current DPCLustX framework outputs one explanation per cluster in the global explanation, aligning with the non-private [7]. It can be generalized to output multiple explanations per cluster for a more comprehensive understanding, but complexity may increase. Extending the framework to efficiently output multiple explanations per cluster in a global explanation is important future work. Second, it would be intriguing to examine the impact of different discretization and binning approaches on the quality and performance of our system. Third, the extension of DPCLustX to different score functions that emphasize different facets of explainability would be an intriguing direction of work. These directions will enhance the usability and interpretability of clustering methods under DP.

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A Theorems and Proofs

Quality functions for histogram-based explanations used in prior work on are closely related to the L_1 distance between histograms, viewed as vectors. This connection will also be used in our analysis. Hence, we now formally define these concepts. A histogram $h_A(D)$ can be viewed as a $|\text{dom}(A)|$ -dimensional vector with value $\text{cnt}_{A=a}(D)$ in its a 'th entry for every $a \in \text{dom}(A)$. The L_1 -norm of a vector $v \in \mathbb{R}^d$, denoted $\|v\|_1$, is defined as $\sum_{i=1}^d |v_i|$. For histogram vectors, we always have $\|h_A(D)\|_1 = |D|$, as the L_1 -norm is simply the sum of all counts. The following is an immediate corollary:

LEMMA A.1. *For a dataset D and a clustering function $f : \text{dom}(R) \rightarrow C$, a cluster label c , and an attribute A ,*

$$\text{Int}_p(D, f, c, A) = \frac{1}{2} \left\| h_A(D_c) - \frac{|D_c|}{|D|} h_A(D) \right\|_1$$

LEMMA A.2. *Let D be a dataset, f a clustering function, A an attribute, and $D_c, D_{c'}$ non-empty clusters such that $|D_c| \leq |D_{c'}|$. The pairwise-diversity score (??) is*

$$d(D, f, c, c', A) = \frac{1}{2} \left\| h_A(D_c) - \frac{|D_c|}{|D_{c'}|} h_A(D_{c'}) \right\|_1$$

Our global quality measures are defined as the average of low-sensitivity functions. Consequently, our analysis frequently uses the following Lemma

LEMMA A.3. *For $i = 1, \dots, m$ let $\alpha_i \in \mathbb{R}$ be a scalar and $f_i : \mathcal{D} \rightarrow \mathbb{R}$ be a function with sensitivity Δ_i . Then, the function $f = \sum_{i=1}^m \alpha_i \cdot f_i$ has sensitivity bounded by $\sum_{i=1}^m |\alpha_i| \Delta_i$.*

PROOF. Let $D \sim D'$ be two neighboring datasets. By the triangle inequality,

$$\begin{aligned} |f(D) - f(D')| &= \left| \sum_{i=1}^m \alpha_i \cdot f_i(D) - \sum_{i=1}^m \alpha_i \cdot f_i(D') \right| \\ &\leq \sum_{i=1}^m |\alpha_i| \cdot |f_i(D) - f_i(D')| \\ &\leq \sum_{i=1}^m |\alpha_i| \Delta_i. \end{aligned}$$

□

A.1 Interestingness

PROPOSITION 4.1. *The sensitivity of dist_{TV} is at least $\frac{1}{2}$ and its range is $[0, 1]$.*

PROOF. Let D be a dataset of size n , and A an attribute. Suppose that for all tuples $t \in D$ it holds that $t[A] = a$ for some $a \in \text{dom}(A)$. That is, $\text{cnt}_{A=a}(D) = n$. Let $D_c \subseteq D$ be a cluster of size 1. In this case $\text{dist}_{TV}(\pi_A(D), \pi_A(D_c)) = 0$, as $\pi_A(D)$ and $\pi_A(D_c)$ define the same distribution (the value a has probability 1).

Now, let $D' = D \cup \{t'\}$ and $D'_c = D_c \cup \{t'\}$ for a tuple t' with $t'[A] = a'$ for $a' \neq a$. We now have

$$\begin{aligned} 2 \cdot \text{dist}_{TV}(\pi_A(D'), \pi_A(D'_c)) &= \left| \frac{\text{cnt}_{A=a}(D')}{|D'|} - \frac{\text{cnt}_{A=a}(D'_c)}{|D'_c|} \right| \\ &\quad + \left| \frac{\text{cnt}_{A=a'}(D')}{|D'|} - \frac{\text{cnt}_{A=a'}(D'_c)}{|D'_c|} \right| \\ &= \left| \frac{n}{n+1} - \frac{1}{2} \right| + \left| \frac{1}{n+1} - \frac{1}{2} \right| \\ &= 1 \end{aligned}$$

15

□

Since previous work has also considered the Jensen-Shannon distance [37] as an interestingness metric, we show that it is highly sensitive as well, making it unsuitable for the DP setting.

Definition A.4 (Jensen-Shannon Divergence [37]). For two distributions p and q over the same domain, their Jensen-Shannon divergence is defined as

$$JSD(p, q) = H\left(\frac{p+q}{2}\right) - \frac{1}{2}H(p) - \frac{1}{2}H(q)$$

where $H(\cdot)$ is the Shannon entropy and $\frac{p+q}{2}$ is the mixture distribution of p and q . The Jensen-Shannon distance, denoted d_{JS} , is the square root of the Jensen-Shannon divergence.

Let $P_A(D)$ and $P_A(D_c)$ be the probability distributions of the values in the columns of $\pi_A(D)$ and $\pi_A(D_c)$, respectively, determined by the relative frequencies of each value. Define $d_{JS}(P_A(D), P_A(D_c))$ as the Jensen-Shannon distance between these distributions. We prove the following:

PROPOSITION A.5. *The sensitivity of d_{JS} is at least $\frac{1}{2}$ and its range is $[0, 1]$.*

PROOF. Let D be a dataset of size n , and A an attribute. Suppose that for all tuples $t \in D$ it holds that $t[A] = a$ for some $a \in \text{dom}(A)$. Let $D_c = \{t\}$ be a cluster of size 1. In this case, $d_{JS}(P_A(D), P_A(D_c)) = 0$, as $P_A(D)$ and $P_A(D_c)$ are the same distribution (the value a has probability 1). Now, let $D' = D \cup \{t'\}$ and $D'_c = D_c \cup \{t'\}$ for a tuple t' with $t'[A] = a'$ where $a' \neq a$. We now have

$$P_A(D') = \begin{cases} a & \text{w.p. } \frac{n}{n+1} \\ a' & \text{w.p. } \frac{1}{n+1} \end{cases}$$

and

$$P_A(D'_c) = \begin{cases} a & \text{w.p. } \frac{1}{2} \\ a' & \text{w.p. } \frac{1}{2} \end{cases}$$

therefore, the mixture distribution satisfies

$$\frac{P_A(D') + P_A(D'_c)}{2} = \begin{cases} a & \text{w.p. } \frac{n}{2(n+1)} + \frac{1}{4} \\ a' & \text{w.p. } \frac{1}{2(n+1)} + \frac{1}{4} \end{cases}$$

Since all three distributions are supported on $\{a, a'\}$, we have

$$JSD(P_A(D'), P_A(D'_c)) = H_b\left(\frac{1}{2(n+1)} + \frac{1}{4}\right) - \frac{1}{2}H_b\left(\frac{1}{n+1}\right) - \frac{1}{2}H_b\left(\frac{1}{2}\right)$$

where H_b is the binary entropy function. Note that by continuity of H_b we have

$$\lim_{n \rightarrow \infty} JSD(P_A(D'), P_A(D'_c)) = H_b\left(\frac{1}{4}\right) - \frac{1}{2}$$

In particular, for sufficiently large n , $JSD(P_A(D'), P_A(D'_c)) > 0.3$, and hence $d_{JS}(P_A(D'), P_A(D'_c)) > \frac{1}{2}$. Therefore, we find that d_{JS} has sensitivity greater than $\frac{1}{2}$. □

Intuitively, since the range of d_{JS} is $[0, 1]$ [37], its sensitivity is relatively high.

PROPOSITION 4.4. *The sensitivity of $\text{Int}_p(D, f, c, A)$ is 1 and its range is $[0, |D_c|]$.*

PROOF. Let $f : \text{dom}(R) \rightarrow C$ be a clustering function, and $c \in C$ be a cluster label. Let $A \in \mathcal{A}$ and $D \sim D'$ be neighboring datasets such that $D' = D \cup \{t\}$. Denote $a = t[A]$, and let $h_a = h_A(t)$ be the $|\text{dom}(A)|$ -dimensional histogram with 1 in its a 'th entry and 0 elsewhere. Let D_c (respectively, D'_c) denote the set of tuples in D (respectively, D') that are mapped to c by the function f , and note that D'_c is either D_c or $D_c \cup \{t\}$.

We first consider the case that $D_c = D'_c$. Applying Lemma A.1 and the triangle inequality, we have

$$\begin{aligned} & |\text{Int}_p(D', f, c, A) - \text{Int}_p(D, f, c, A)| \\ &= \frac{1}{2} \left\| h_A(D_c) - \frac{|D_c|}{|D'|} h_A(D') \right\|_1 - \left\| h_A(D_c) - \frac{|D_c|}{|D|} h_A(D) \right\|_1 \\ &\leq \frac{1}{2} \left\| \frac{|D_c|}{|D|} h_A(D) - \frac{|D_c|}{|D'|} h_A(D') \right\|_1 \end{aligned}$$

Substituting $h_A(D') = h_A(D) + h_a$ into the inequality above and applying the triangle inequality again, we obtain

$$\begin{aligned} &= \frac{1}{2} \left\| \frac{|D_c|}{|D|+1} h_a - \frac{|D_c|}{|D|(|D|+1)} h_A(D) \right\|_1 \\ &\leq \frac{1}{2} \left(\|h_a\|_1 + \frac{1}{|D|} \|h_A(D)\|_1 \right) \end{aligned}$$

where for the equality, we used the fact that $|D'| = |D| + 1$, which implies $\frac{|D_c|}{|D|} - \frac{|D_c|}{|D'|} = \frac{|D_c|}{|D|(|D|+1)}$. For the inequality, we also bound $|D_c| \leq |D| + 1$. Next, we substitute $\|h_a\|_1 = 1$ and $\|h_A(D)\|_1 = |D|$ to obtain

$$= \frac{1}{2} \left(1 + \frac{|D|}{|D|} \right) = 1$$

Now, consider the case that $D'_c = D_c \cup \{t\}$. Applying the triangle inequality, we have

$$\begin{aligned} & |\text{Int}_p(D', f, c, A) - \text{Int}_p(D, f, c, A)| \\ &\leq \frac{1}{2} \left\| h_A(D'_c) - \frac{|D'_c|}{|D'|} h_A(D') - h_A(D_c) + \frac{|D_c|}{|D|} h_A(D) \right\|_1. \end{aligned}$$

Substituting $h_A(D') = h_A(D) + h_a$ and $h_A(D'_c) = h_A(D_c) + h_a$, and rearranging, we obtain

$$\begin{aligned} &= \frac{1}{2} \left\| h_a - \frac{|D'_c|}{|D'|} h_A(D) - \frac{|D'_c|}{|D'|} h_a + \frac{|D_c|}{|D|} h_A(D) \right\|_1 \\ &= \frac{1}{2} \left\| \left(1 - \frac{|D'_c|}{|D'|} \right) h_a - \left(\frac{|D'_c|}{|D'|} - \frac{|D_c|}{|D|} \right) h_A(D) \right\|_1 \end{aligned}$$

Recall that $|D'_c| = |D_c| + 1$ and $|D'| = |D| + 1$. Therefore, $|D'_c|/|D'| - |D_c|/|D| = (|D| - |D_c|)/(|D|(|D|+1))$, which is non-negative, as $|D_c| \leq |D|$. Since $|D'_c| \leq |D'|$ holds as well, we have

$$\begin{aligned} &\leq \frac{1}{2} \left(1 - \frac{|D'_c|}{|D'|} \right) \|h_a\|_1 + \frac{1}{2} \left(\frac{|D| - |D_c|}{|D|(|D|+1)} \right) \|h_A(D)\|_1 \\ &\leq \frac{1}{2} \left(1 - \frac{|D'_c|}{|D'|} \right) \|h_a\|_1 + \frac{1}{2} \left(\frac{1}{|D|+1} \right) \|h_A(D)\|_1. \end{aligned}$$

Since $\|h_a\|_1 = 1$ and $\|h_A(D)\|_1 = |D|$, we conclude

$$\begin{aligned} &= \frac{1}{2} \left(1 - \frac{|D'_c|}{|D'|} \right) + \frac{1}{2} \left(\frac{1}{|D|+1} \right) |D| \\ &\leq \frac{1}{2} + \frac{1}{2} = 1. \end{aligned}$$

For the range upper bound, we use Lemma A.1 to obtain

$$\begin{aligned} \text{Int}_p(D, f, c, A) &= \frac{1}{2} \left\| h_A(D_c) - \frac{|D_c|}{|D|} h_A(D) \right\|_1 \\ &\leq \frac{1}{2} \left(\|h_A(D_c)\|_1 + \frac{|D_c|}{|D|} \|h_A(D)\|_1 \right) \\ &= |D_c| \end{aligned}$$

□

A.2 Sufficiency

We revisit the notion of sufficiency from prior work and provide its sensitivity analysis.

$$m_{\mathcal{AC}}^s(t) = \frac{\sum_{t' \in D} \mathbf{1}_{\{f(t')=f(t)\}} \cdot r(t', A_c)}{\sum_{t' \in D} r(t', A_c)} \quad (2)$$

where we denote $c = f(t)$ and $A_c = \mathcal{AC}(c)$. Observe that $m_{\mathcal{AC}}^s(t)$ attains its maximum 1 when the value $t[A_c]$ appears only in the cluster D_c , capturing the notion of maximal sufficiency. To measure the *global* sufficiency of $\mathcal{AC}$, [7] averages the local sufficiency across all tuples:

$$\text{Suf}(D, f, \mathcal{AC}) = \frac{1}{|D|} \sum_{t \in D} m_{\mathcal{AC}}^s(t). \quad (3)$$

In the unbounded-DP variant, where the dataset size is not fixed and which we adopt in this work, this function exhibits high sensitivity relative to its range

PROPOSITION 4.5. *The sensitivity of $\text{Suf}(D, f, \mathcal{AC})$ is at least $\frac{1}{2}$ and its range is $[0, 1]$.*

PROOF. We use the equality from item (1) of Proposition 4.7. Consider a dataset $D = \{t_1, t_2\}$ and two clusters $D_i = \{t_i\}$ for $i = 1, 2$. Let $\mathcal{AC}$ be an attribute combination, and denote $A_i = \mathcal{AC}(i)$ for $i = 1, 2$.

Assume that for all $i \in \{1, 2\}$ $t_1[A_i] \neq t_2[A_i]$. In this case $\text{Suf}_p(D, f, i, A_i) = 1$, as the value $t_i[A_i]$ appears only inside the cluster D_i . Hence,

$$\text{Suf}(D, \mathcal{AC}, f) = \frac{1}{|D|} \sum_{c \in C} \text{Suf}_p(D, f, c, \mathcal{AC}(c)) = \frac{1}{2} \cdot (1 + 1) = 1.$$

Now, suppose a tuple t_3 is added to the cluster D_2 . That is, $D' = \{t_1, t_2, t_3\}$, $D'_1 = \{t_1\}$ and $D'_2 = \{t_2, t_3\}$. Suppose further that $t_3[A_1] = t_1[A_1]$, and $t_3[A_2] = t_1[A_2]$. In particular $t_3[A_2] \neq t_2[A_2]$.

In this case

$$\begin{aligned} \text{Suf}_p(D', f, 1, A_1) &= \frac{1}{|D'_1|} \sum_{a \in \text{dom}_{D'_1}(A_1)} \frac{(\text{cnt}_{A_1=a}(D'_1))^2}{\text{cnt}_{A_1=a}(D')} \\ &= \frac{1}{1} \cdot \frac{1}{2} = \frac{1}{2}. \end{aligned}$$

Moreover,

$$\begin{aligned} \text{Suf}_p(D', f, 2, A_2) &= \frac{1}{|D'_2|} \sum_{a \in \text{dom}_{D'_2}(A_2)} \frac{(\text{cnt}_{A_2=a}(D'_2))^2}{\text{cnt}_{A_2=a}(D')} \\ &= \frac{1}{2} \left(\frac{1}{1} + \frac{1}{2} \right) = \frac{3}{4}. \end{aligned}$$

Hence, we find that

$$\begin{aligned} \text{Suf}(D', \mathcal{AC}, f) &= \frac{1}{|D'|} \sum_{c \in \{1,2\}} \text{Suf}_p(D', f, c, \mathcal{AC}(c)) = \frac{1}{3} \cdot \left(\frac{1}{2} + \frac{3}{4} \right) \\ &< \frac{1}{2}. \end{aligned}$$

□

PROPOSITION 4.7. For an attribute combination $\mathcal{AC}$,

(1) The following equality holds:

$$|D| \cdot \text{Suf}(D, f, \mathcal{AC}) = \sum_{c \in C} \text{Suf}_p(D, f, c, \mathcal{AC}(c))$$

(2) The sensitivity of $\text{Suf}_p(D, f, c, A)$ is 1 and its range is $[0, |D_c|]$.

PROOF OF (1). Let $\mathcal{AC}$ be an attribute combination and $t \in D$. Denote $c = f(t)$, and $A_c = \mathcal{AC}(c)$. Observe that the right-hand-side of Equation (2) depends only on the cluster assignment c , therefore it is equal across all tuples t of the same cluster. Thus, plugging in the definition of $r(t', A)$,

$$\sum_{t \in D} m_E^s(t) = \sum_{c \in C} |D_c| \cdot \frac{\sum_{t \in D_c} \Pr[D_c | A_c = t[A_c]]}{\sum_{t \in D} \Pr[D_c | A_c = t[A_c]]}. \quad (4)$$

Let us consider each cluster separately. Using the definition for $\Pr[D_c | A = t[A]]$, we have

$$\frac{\sum_{t \in D_c} \Pr[D_c | A_c = t[A_c]]}{\sum_{t \in D} \Pr[D_c | A_c = t[A_c]]} = \frac{\sum_{t \in D_c} \frac{\text{cnt}_{A_c=t[A_c]}(D_c)}{\text{cnt}_{A_c=t[A_c]}(D)}}{\sum_{t \in D} \frac{\text{cnt}_{A_c=t[A_c]}(D_c)}{\text{cnt}_{A_c=t[A_c]}(D)}}.$$

Now, observe that each value $\text{cnt}_{A_c=a}(D_c) / \text{cnt}_{A_c=a}(D)$ appears in the numerator exactly $\text{cnt}_{A_c=a}(D_c)$ times, and $\text{cnt}_{A_c=a}(D)$ times in the denominator. Hence, by changing the order of summation

$$\begin{aligned} &= \frac{\sum_{a \in \text{dom}_{D_c}(A_c)} \text{cnt}_{A_c=a}(D_c) \cdot \frac{\text{cnt}_{A_c=a}(D_c)}{\text{cnt}_{A_c=a}(D)}}{\sum_{a \in \text{dom}_D(A_c)} \text{cnt}_{A_c=a}(D) \cdot \frac{\text{cnt}_{A_c=a}(D_c)}{\text{cnt}_{A_c=a}(D)}} \\ &= \frac{\sum_{a \in \text{dom}_{D_c}(A_c)} \frac{(\text{cnt}_{A_c=a}(D_c))^2}{\text{cnt}_{A_c=a}(D)}}{\sum_{a \in \text{dom}_D(A_c)} \text{cnt}_{A_c=a}(D_c)} \end{aligned}$$

Since the denominator equals $|D_c|$, the following equality holds:

$$\begin{aligned} &= \frac{1}{|D_c|} \sum_{a \in \text{dom}_{D_c}(A_c)} \frac{(\text{cnt}_{A_c=a}(D_c))^2}{\text{cnt}_{A_c=a}(D)} \\ &= \frac{1}{|D_c|} \text{Suf}_p(D, f, c, A). \end{aligned}$$

The proposition follows by substituting the equality into Equation (4). □

To prove item (2) of Proposition 4.7, we introduce the following to lemmas.

LEMMA A.6. The range of $\text{Suf}_p(D, f, c, A)$ is $[0, |D_c|]$.

PROOF. We have

$$\begin{aligned} \text{Suf}_p(D, f, c, A) &= \sum_{a \in \text{dom}_{D_c}(A)} \frac{\text{cnt}_{A=a}(D_c)^2}{\text{cnt}_{A=a}(D)} \\ &\leq \sum_{a \in \text{dom}_{D_c}(A)} \text{cnt}_{A=a}(D_c) \\ &= |D_c| \end{aligned}$$

where the inequality holds since $\text{cnt}_{A=a}(D_c) \leq \text{cnt}_{A=a}(D)$. □

LEMMA A.7. Let $a, b \in \mathbb{R}$ such that $b > 0$ and $0 \leq a \leq b$. Then

$$\begin{aligned} (i) \quad &\left| \frac{a^2}{b} - \frac{(a+1)^2}{b+1} \right| \leq 1, \\ (ii) \quad &\left| \frac{a^2}{b} - \frac{a^2}{b+1} \right| \leq 1 \end{aligned}$$

PROOF. For item (i), observe that our assumption $a \leq b$ implies that $\frac{(a+1)^2}{b+1} \geq \frac{a^2}{b}$. Indeed,

$$\begin{aligned} \frac{a^2}{b} &= \frac{a^2(b+1)}{b(b+1)} \\ &= \frac{a^2b + a^2}{b(b+1)} \leq \frac{a^2b + 2ab + b}{b(b+1)} = \frac{b(a^2 + 2a + 1)}{b(b+1)} = \frac{(a+1)^2}{b+1}. \end{aligned}$$

where the in the inequality we used $a^2 \leq ab \leq 2ab + b$. Now, we have

$$\begin{aligned} \frac{(a+1)^2}{b+1} - \frac{a^2}{b} &= \frac{b(a+1)^2 - a^2(b+1)}{b(b+1)} \\ &= \frac{2ab + b - a^2}{b(b+1)} \leq \frac{b^2 + b}{b(b+1)} = 1 \end{aligned}$$

where the inequality follows from $2ab - a^2 \leq b^2$ which holds for any $a, b \in \mathbb{R}$.

For (ii), note that

$$\frac{a^2}{b} - \frac{a^2}{b+1} = \frac{a^2}{b(b+1)} < 1$$

as $a \leq b$. □

PROOF OF PROPOSITION 4.7 ITEM (2). Let $f : \text{dom}(R) \rightarrow C$ be a clustering function, and $c \in C$ be a cluster index. Fix an attribute A . Let $D \sim D'$ be two neighboring datasets such that $D' = D \cup \{t\}$. Let D_c (respectively, D'_c) denote the set of tuples in D (respectively, D') that are mapped to c by the function f , and note that D'_c is either D_c or $D_c \cup \{t\}$. First, note that if D_c is empty,

$$|\text{Suf}_p(D, f, c, A) - \text{Suf}_p(D', f, c, A)| \leq 1.$$

Indeed, in this case $|D'_c| \leq 1$, and by Lemma A.6 we have $\text{Suf}_p(D, f, c, A) = 0$ and $\text{Suf}_p(D', f, c, A) \leq 1$.

Second, if $\text{cnt}_{A=t[A]}(D) = 0$, then clearly $\text{cnt}_{A=t[A]}(D_c) = 0$ and $\text{cnt}_{A=t[A]}(D'_c) \leq 1$. In this case

$$\begin{aligned} & |\text{Suf}_p(D, f, c, A) - \text{Suf}_p(D', f, c, A)| \\ &= \left| \sum_{a \in \text{dom}_{D_c}(A)} \frac{\text{cnt}_{A=a}(D_c)^2}{\text{cnt}_{A=a}(D)} - \sum_{a \in \text{dom}_{D'_c}(A)} \frac{\text{cnt}_{A=a}(D'_c)^2}{\text{cnt}_{A=a}(D')} \right| \\ &= \frac{\text{cnt}_{A=t[A]}(D'_c)^2}{\text{cnt}_{A=t[A]}(D')} \\ &\leq 1. \end{aligned}$$

where the second equality holds because all summands are identical in both sums, except for the term corresponding to $a = t[A]$, which appears only in the second sum. The inequality holds since the numerator is at most 1.

Hence, we assume that D_c is not empty and that $\text{cnt}_{A=t[A]}(D) > 0$. We consider two cases according to whether or not t is added to D_c .

First assume that $D'_c = D_c$. We have

$$\begin{aligned} & |\text{Suf}_p(D, f, c, A) - \text{Suf}_p(D', f, c, A)| \\ &= \left| \frac{\text{cnt}_{A=t[A]}(D_c)^2}{\text{cnt}_{A=t[A]}(D)} - \frac{\text{cnt}_{A=t[A]}(D_c)^2}{\text{cnt}_{A=t[A]}(D) + 1} \right| \leq 1 \end{aligned}$$

where in the equality we have used that $\text{cnt}_{A=a}(D_c) = \text{cnt}_{A=a}(D'_c)$ for all a since $D'_c = D_c$. We also used that the only summand in the definition of Suf_p that changes is the one corresponding to $t[A]$. The inequality uses item (i) of Lemma A.7.

If $D'_c = D_c \cup \{t\}$, we use item (ii) of Lemma A.7 to obtain

$$\begin{aligned} & |\text{Suf}_p(D, f, c, A) - \text{Suf}_p(D', f, c, A)| \\ &= \left| \frac{\text{cnt}_{A=t[A]}(D_c)^2}{\text{cnt}_{A=t[A]}(D)} - \frac{(\text{cnt}_{A=t[A]}(D_c) + 1)^2}{\text{cnt}_{A=t[A]}(D) + 1} \right| \leq 1. \end{aligned}$$

□

A.3 Diversity

We begin with a sensitivity analysis of the diversity function from prior work, and proceed with the analysis of our low sensitivity diversity measure Definition 4.9.

The diversity from [7] is defined as follows. Let $\mathcal{AC} : C \rightarrow \mathcal{A}$ be an attribute combinations. For an attribute A , let $\text{ExpBy}(\mathcal{AC}, A) = \mathcal{AC}^{-1}(\{A\})$. That is, the set of cluster labels assigned to A by $\mathcal{AC}$ (which can also be empty).

For a finite set S , let $\text{Perm}(S)$ denote its set of permutations, containing all bijections $p : \{1, \dots, |S|\} \rightarrow S$. For a permutation $p \in \text{Perm}(\text{ExpBy}(\mathcal{AC}, A))$, its diversity is defined as [7]:

$$\text{PermDiv}_A(p) = \sum_{i=1}^{|\text{ExpBy}(\mathcal{AC}, A)|} \min_{j < i} d_{TV}(\pi_A(D_{p(i)}), \pi_A(D_{p(j)}))$$

and if $|\text{ExpBy}(\mathcal{AC}, A)| = 1$, $\text{PermDiv}_A(p)$ is set as 1.

Now, the diversity measure is defined as⁶

$$\text{Div}(D, f, \mathcal{AC}) = \sum_{A \in \mathcal{A}} \sum_{p \in \text{Perm}(\text{ExpBy}(\mathcal{AC}, A))} \frac{\text{PermDiv}_A(p)}{|\text{ExpBy}(\mathcal{AC}, A)|!}$$

⁶to obtain a value between 0 and 1, this function can be normalized by the number of clusters $|C|$.

PROPOSITION A.8. *The sensitivity of $\text{Div}(D, f, \mathcal{AC})$ is at least $\frac{1}{2}$ and its range is $[0, |C|]$.*

Note that since the number of clusters $|C|$ is typically a small constant, the sensitivity is relatively high compared to the range.

PROOF. Let D be a dataset of size n and $f : \text{dom}(R) \rightarrow C$ be a clustering function such that $D_1 = \{t_1\}$ is a cluster of size 1. Let A be an attribute and suppose that there exists $a \in \text{dom}(A)$ such that $t[A] = a$ for all tuples $t \in D$. Let $\mathcal{AC}$ be an attribute combination mapping all cluster labels to A .

In this case, we have $\text{ExpBy}(\mathcal{AC}, A) = C$. Moreover, for any $c, c' \in C$ we have that $d_{TV}(\pi_A(D_c), \pi_A(D_{c'})) = 0$, as the distributions of values of the column A is identical among all clusters. Therefore, for any permutation $p \in \text{Perm}(C)$, we find that $\text{PermDiv}_A(p) = 0$. Hence, $\text{Div}(D, f, \mathcal{AC}) = 0$.

Now, suppose a new tuple is added to cluster 1. Denote $D' = D \cup \{t_2\}$ and $D'_1 = \{t_1, t_2\}$. Suppose further that $t_2[A] = a'$ for $a' \neq a$. Now, for every $c \neq 1$ we have

$$d_{TV}(\pi_A(D_1), \pi_A(D_c)) = \frac{1}{2} \quad (5)$$

and for every $c, c' \in C \setminus \{1\}$ we have

$$d_{TV}(\pi_A(D_1), \pi_A(D_c)) = 0 \quad (6)$$

as for other clusters the distribution of values is unchanged. Hence, for any permutation $p \in \text{Perm}(C)$, we find that

$$\text{PermDiv}_A(p) = \sum_{i=1}^{|C|} \min_{j < i} d_{TV}(\pi_A(D_{p(i)}), \pi_A(D_{p(j)})) = \frac{1}{2},$$

where we used that exactly one of the summands has the form (5) and the rest have the form (6). Therefore, we conclude that $\text{Div}(D, f, \mathcal{AC}) = 1/2$.

For the range bound, observe that $\text{Div}(D, f, \mathcal{AC})$ is maximized when all pairwise distances equal 1. This can occur, for instance, when $\mathcal{AC}$ maps each cluster to a distinct attribute. Alternatively, when any two clusters assigned with the same attribute A satisfy $d_{TV}(\pi_A(D_1), \pi_A(D_c)) = 1$. Assuming this is the case, we have

$$\text{PermDiv}_A(p) = \sum_{i=1}^{|\text{ExpBy}(\mathcal{AC}, A)|} 1 = |\text{ExpBy}(\mathcal{AC}, A)|.$$

Thus,

$$\begin{aligned} \text{Div}(D, f, \mathcal{AC}) &= \sum_{A \in \mathcal{A}} \sum_{p \in \text{Perm}(\text{ExpBy}(\mathcal{AC}, A))} \frac{\text{PermDiv}_A(p)}{|\text{ExpBy}(\mathcal{AC}, A)|!} \\ &= \sum_{A \in \mathcal{A}} \sum_{p \in \text{Perm}(\text{ExpBy}(\mathcal{AC}, A))} \frac{|\text{ExpBy}(\mathcal{AC}, A)|}{|\text{ExpBy}(\mathcal{AC}, A)|!} \\ &= \sum_{A \in \mathcal{A}} |\text{ExpBy}(\mathcal{AC}, A)| \\ &= |C|. \end{aligned}$$

□

We now proceed with the sensitivity analysis of our diversity measure (Definition 4.9).

LEMMA A.9. *For two clusters $D_c, D_{c'} \subseteq D$ and attributes $A_c, A_{c'}$, it holds that $d(D, f, c, c', A_c, A_{c'}) \in [0, \min\{|D_c|, |D_{c'}|\}]$*

PROOF. If one of the two clusters is empty, or if $A_c \neq A_{c'}$, then the claim trivially holds. Hence, we assume that both are not empty and that $A_c = A_{c'} = A$. In this case

$$\begin{aligned} & \frac{1}{2} \sum_{a \in \text{dom}(A)} \left| \frac{\text{cnt}_{A=a}(D_c)}{\max\{|D_c|, 1\}} - \frac{\text{cnt}_{A=a}(D_{c'})}{\max\{|D_{c'}|, 1\}} \right| \\ & \leq \frac{1}{2} \sum_{a \in \text{dom}(A)} \left| \frac{\text{cnt}_{A=a}(D_c)}{|D_c|} \right| + \frac{1}{2} \sum_{a \in \text{dom}(A)} \left| \frac{\text{cnt}_{A=a}(D_{c'})}{|D_{c'}|} \right| \\ & = \frac{1}{2} + \frac{1}{2} = 1 \end{aligned}$$

and therefore $d(D, f, c, c', A_c, A_{c'}) \leq \min\{|D_c|, |D_{c'}|\}$. $\square$

PROPOSITION A.10. d has sensitivity 1.

The proof is similar to that of Proposition 4.4, with the main difference being that we now consider the distance between histograms of disjoint subsets of the data, instead of one subset and the entire dataset.

PROOF. Let $f : \text{dom}(R) \rightarrow C$ be a clustering function, and $c, c' \in C$ be two cluster labels. Fix attributes $A_c, A_{c'}$, and let $D \sim D'$ be neighboring datasets such that $D' = D \cup \{t\}$.

Let D_c and $D_{c'}$ (respectively, D'_c and $D'_{c'}$) denote the sets of tuples in D (respectively, D') that are mapped to c and c' by the function f .

First, observe that when $A_c \neq A_{c'}$, we have $d(D, f, c, c', A_c, A_{c'}) = \min\{|D_c|, |D_{c'}|\}$, which changes by at most 1 when we add or remove one tuple from D . Hence, we proceed with the assumption that $A_c = A_{c'} = A$. To simplify notation, we let $d(D, c, c', A) = d(D, f, c, c', A, A)$. If t has not been added to either cluster, i.e., $D_c = D'_c$ and $D_{c'} = D'_{c'}$, then clearly

$$|d(D, c, c', A) - d(D', c, c', A)| = 0.$$

Otherwise, consider the case that t has been added to one of the clusters. Without loss of generality, suppose that $D'_{c'} = D_{c'} \cup \{t\}$ and $D'_c = D_c$. If either of D_c or $D_{c'}$ is empty, then by Lemma A.9, we have $d(D_c, D_{c'}, A) = 0$, and $d(D_c, D'_{c'}, A) \leq 1$, and so

$$|d(D, c, c', A) - d(D', c, c', A)| \leq 1.$$

Hence, we proceed with the assumption that both are not empty. Denote $a = t[A]$, and let $h_a = h_A(t)$ be the $|\text{dom}(A)|$ -dimensional histogram with 1 in its a 'th entry and 0 elsewhere. First, consider the case that $|D_c| \leq |D_{c'}|$. Recalling that $|D'_{c'}| = |D_{c'}| + 1$, we have

$$\begin{aligned} & |d(D, c, c', A) - d(D', c, c', A)| \\ & = \frac{1}{2} \left\| h_A(D_c) - \frac{|D_c|}{|D_{c'}|} h_A(D_{c'}) \right\|_1 - \left\| h_A(D_c) + \frac{|D_c|}{|D_{c'}| + 1} h_A(D'_{c'}) \right\|_1 \end{aligned}$$

By the triangle inequality,

$$\leq \frac{1}{2} \left\| h_A(D_c) - \frac{|D_c|}{|D_{c'}|} h_A(D_{c'}) - h_A(D_c) + \frac{|D_c|}{|D_{c'}| + 1} h_A(D'_{c'}) \right\|_1$$

Substituting $h_A(D'_{c'}) = h_A(D_{c'}) + h_a$ into the inequality above, we obtain

$$= \frac{1}{2} \left\| \frac{|D_c|}{|D_{c'}| + 1} h_a - \left(\frac{|D_c|}{|D_{c'}|} - \frac{|D_c|}{|D_{c'}| + 1} \right) h_A(D_{c'}) \right\|_1$$

Applying the triangle inequality and noting that $\frac{|D_c|}{|D_{c'}|} - \frac{|D_c|}{|D_{c'}| + 1} = \frac{|D_c|}{|D_{c'}|(|D_{c'}| + 1)}$, we have

$$\leq \frac{1}{2} \left(\|h_a\|_1 + \frac{|D_c|}{|D_{c'}|(|D_{c'}| + 1)} \|h_A(D_{c'})\|_1 \right)$$

substituting $\|h_a\|_1 = 1$ and $\|h_A(D_{c'})\|_1 = |D_{c'}|$,

$$= \frac{1}{2} \left(1 + \frac{|D_c| \cdot |D_{c'}|}{|D_{c'}|(|D_{c'}| + 1)} \right) \leq 1$$

where we have also used our assumption that $|D_c| \leq |D_{c'}|$.

Now, consider the case that $|D_c| > |D_{c'}|$, and therefore also $|D_c| \geq |D'_{c'}| = |D_{c'}| + 1$. We have

$$\begin{aligned} & |d(D, c, c', A) - d(D', c, c', A)| \\ & = \frac{1}{2} \left\| \frac{|D_{c'}|}{|D_c|} h_A(D_c) - h_A(D_{c'}) \right\|_1 - \left\| \frac{|D_{c'}| + 1}{|D_c|} h_A(D_c) - h_A(D'_{c'}) \right\|_1 \end{aligned}$$

By the triangle inequality,

$$\leq \frac{1}{2} \left\| \frac{|D_{c'}|}{|D_c|} h_A(D_c) - h_A(D_{c'}) - \frac{|D_{c'}| + 1}{|D_c|} h_A(D_c) + h_A(D'_{c'}) \right\|_1$$

Substituting $h_A(D'_{c'}) = h_A(D_{c'}) + h_a$ into the inequality above and rearranging, we obtain

$$\leq \frac{1}{2} \left\| \frac{-1}{|D_c|} h_A(D_c) + h_a \right\|_1$$

Next, we once again use $\|h_a\|_1 = 1$ and $\|h_A(D_c)\|_1 = |D_c|$

$$\begin{aligned} & \leq \frac{1}{2} \left(\frac{1}{|D_c|} \|h_A(D_c)\|_1 + \|h_a\|_1 \right) \\ & = 1 \end{aligned}$$

$\square$

PROPOSITION 4.10. The sensitivity of Div_p is bounded by 1. Moreover, its range is $[0, R_{\text{Div}}]$ where

$$R_{\text{Div}} = \frac{1}{\binom{|C|}{2}} \sum_{i=1}^{|C|} (|C| - i) |D_{c_i}|$$

is a weighted average of the cluster sizes, and $|D_{c_i}| \leq |D_{c_{i+1}}|$.

PROOF. The sensitivity bound follows from Proposition A.10 using Lemma A.3, as Div_p is a convex combination of sensitivity-1 functions. For the range bound, note that $d(D, f, c, c', A_c, A_{c'})$ attains its upper-bound (Lemma A.9), in particular, when $A_c \neq A_{c'}$. Hence, Let $\mathcal{AC}$ be an attribute combination assigning a different attribute for each cluster. It follows that Div_p is maximized, attaining value

$$\frac{1}{\binom{|C|}{2}} \sum_{\{c, c'\} \subseteq C} \min\{|D_c|, |D_{c'}|\}. \quad (7)$$

Now, let $|D_{c_1}| \leq \dots \leq |D_{c_{|C|}}|$ be an ordering of the clusters by increasing size. For each $i = 1, \dots, |C|$, the value $|D_{c_i}|$ appears as a summand in (7) $|C| - i$ times, each for a coupling of c_i with c_j for $j < i$. By changing the order of summation, we obtain that (7) equals

$$\frac{1}{\binom{|C|}{2}} \sum_{i=1}^{|C|} (|C| - i) |D_{c_i}|$$

$\square$

A.4 Combining All Quality Functions

PROPOSITION 4.12. $\text{Score}_\gamma(D, f, c, A)$ has sensitivity bounded by 1 and its range is $[0, |D_c|]$.

PROOF. The proposition follows directly from Lemma A.3 and the sensitivity bounds of the sufficiency and interestingness functions (Proposition 4.4, Proposition 4.7), as Score_γ is a convex combination of the two. $\square$

PROPOSITION 4.14. GScore_λ has sensitivity bounded by 1. Moreover, its range is $[0, R_{\text{GScore}_\lambda}]$ where

$$R_{\text{GScore}_\lambda} = (\lambda_{\text{Int}} + \lambda_{\text{Suf}}) \cdot \frac{1}{|C|} \sum_{c \in C} |D_c| + \lambda_{\text{Div}} \cdot R_{\text{Div}}$$

is a weighted average of the cluster sizes, and R_{Div} is defined as in Proposition 4.10.

PROOF. The proposition follows directly from Lemma A.3 and the sensitivity bounds of the sufficiency, interestingness and diversity functions (Proposition 4.4, Proposition 4.7, Proposition 4.10), as GScore_λ is a convex combination of the three. For the range bounds, note that $\frac{1}{|C|} \sum_{c \in C} \text{Suf}_p(D, f, c, A)$ and $\frac{1}{|C|} \sum_{c \in C} \text{Suf}_p(D, f, c, A)$ both have range $[0, \frac{1}{|C|} \sum_{c \in C} |D_c|]$. The range upper bound for $\text{Div}_p, R_{\text{Div}_p}$, is given by Proposition 4.10. $\square$

A.5 Proof of Proposition 5.1

PROPOSITION 5.1. Given a clustering function f , a set of attributes $\mathcal{A}$, a privacy parameter $\epsilon_{\text{CandList}}$, non-negative hyperparameters $\gamma_{\text{Int}}, \gamma_{\text{Suf}}$ that sum to 1, and a size parameter k , the following holds:

- (1) Algorithm 1 satisfies $\epsilon_{\text{CandList}}\text{-DP}$.
- (2) For each $c \in C$, denote by $\text{OPT}_c^{(\ell)}$ the ℓ -th highest (true) score, and by $A_c^{(\ell)}$ the ℓ -th explaining attribute selected by Algorithm 1 to S_c . For all c and for all $\ell = 1, 2, \dots, k$, we have

$$\Pr \left[\text{Score}(A_c^{(\ell)}) \leq \text{OPT}_c^{(\ell)} - \frac{2|C| \cdot k}{\epsilon_{\text{CandList}}} (\ln |\mathcal{A}| + t) \right] \leq e^{-t}.$$

PROOF. (1) **Privacy.** For each cluster, the distribution of the selected top- k is equivalent to iteratively applying k exponential mechanisms [13], where each satisfies $\epsilon = \epsilon_{\text{CandList}} / (|C| \cdot k)$. Overall, we have $|C|$ applications of Top- k , hence the output distribution is equivalent to $|C| \cdot k$ exponential mechanisms. by sequential composition (Proposition 2.8), Algorithm 1 satisfies overall $\epsilon_{\text{CandList}}\text{-DP}$.

(2) **Utility Bound.** Let $c \in C$, and $\mathcal{A}_\ell$ be the set of remaining attributes after the $(\ell - 1)$ -th selection for this cluster. The distribution over the selected sequence $A_c^{(1)}, \dots, A_c^{(k)}$ is equal to applying k exponential mechanisms [13] where $A_c^{(\ell)}$ is selected from $\mathcal{A}_\ell$. Note that we always have $\max_{A \in \mathcal{A}_\ell} \text{Score}(A) \geq \text{OPT}_c^{(\ell)}$. Hence, for $\epsilon = \epsilon_{\text{CandList}} / (|C| \cdot k)$, we apply the utility theorem of exponential mechanism (Theorem 3.11 in [17]) to obtain

$$\begin{aligned} \Pr \left[\text{Score}(A_c^{(\ell)}) \leq \text{OPT}_c^{(\ell)} - \frac{2}{\epsilon} (\ln |\mathcal{A}| + t) \right] \\ \leq \Pr \left[\text{Score}(A_c^{(\ell)}) \leq \max_{A \in \mathcal{A}_\ell} \text{Score}(A) - \frac{2}{\epsilon} (\ln |\mathcal{A}_\ell| + t) \right] \leq e^{-t} \end{aligned}$$

where we have used that $\Delta_{\text{Score}} \leq 1$. $\square$

B Providing Multiple Explanations per Cluster

In this section we show how DPCLustX can be extended to output multiple histograms per cluster in a global explanation. To this end, we extend the definition of an *attribute combination* to a mapping $\mathcal{AC} : C \rightarrow \{S \subseteq \mathcal{A} \mid |S| = \ell\}$, mapping each cluster label to a set of ℓ attributes. Our goal is thus to find a high-quality attribute combination such that the histograms of the corresponding attributes form a high-quality HBE, where the global explanation contains ℓ histograms per cluster. This requires extending the global score function Definition 4.13 to attribute combinations with larger outputs.

We measure the overall diversity as follows. Let

$$\text{Cand}(\mathcal{AC}) = \{(c, A) \mid c \in C, A \in \mathcal{AC}(c)\}$$

$$\text{Div}_\ell(D, f, \mathcal{AC}) = \frac{1}{\binom{|\text{Cand}(\mathcal{AC})|}{2}} \sum_{(c, A), (c', A')} d(D, f, c, c', A, A')$$

where the sum is over all different pairs $\{(c, A), (c', A')\} \subseteq \text{Cand}(\mathcal{AC})$. Note that the definition coincides with Definition 4.9 when $\ell = 1$.

The sufficiency and interestingness in the global score function Definition 4.13 were averages across all candidates. With the new definition, they remain averages, but are now taken over a larger set of candidates. Overall, we have:

$$\begin{aligned} \text{GScore}_\lambda(D, f, \mathcal{AC}) &= \lambda_{\text{Int}} \cdot \text{Int}_\ell(D, f, \mathcal{AC}) + \lambda_{\text{Suf}} \cdot \text{Suf}_\ell(D, f, \mathcal{AC}) \\ &\quad + \lambda_{\text{Div}} \cdot \text{Div}_\ell(D, f, \mathcal{AC}) \end{aligned}$$

where we extend

$$\text{Int}_\ell(D, f, \mathcal{AC}) = \frac{1}{|\text{Cand}(\mathcal{AC})|} \sum_{(c, A) \in \text{Cand}(\mathcal{AC})} \text{Int}_p(D, f, c, A)$$

and

$$\text{Suf}_p(D, f, \mathcal{AC}) = \frac{1}{|\text{Cand}(\mathcal{AC})|} \sum_{(c, A) \in \text{Cand}(\mathcal{AC})} \text{Suf}_p(D, f, c, A).$$

Note that the definition coincides with Definition 4.13 when $\ell = 1$. The sensitivity of this function is bounded by 1, as it remains a convex combination of sensitivity-1 functions, with an analysis analogous to that of Proposition 4.14.

Stage-1 of DPCLustX is unchanged, as its purpose is to narrow the search space to high-quality candidates for each cluster, which then form the pool for a global explanation. For Stage-2, we may use the exponential mechanism to select a high-scoring attribute combination $\mathcal{AC} : C \rightarrow \{S \subseteq \mathcal{A} \mid |S| = \ell\}$ with respect to the extended low sensitivity global score. Finally, As in the case of $\ell = 1$, noisy histograms are generated only for the $|C| \times \ell$ selected attributes.

However, the drawback of this approach is that the set of all possible attribute combinations, after the filtering performed in Stage-1, now has a size of $\binom{k}{\ell}^{|C|}$. This size can be large, and computing the global score for all attribute combinations may require significant computational time. While the utility guarantee of Algorithm 1 remains the same, the candidate set provided to the exponential mechanism in Algorithm 2 now has a size of $\binom{k}{\ell}^{|C|}$ instead of $k^{|C|}$, resulting in a larger additive error term of the EM utility bound. Indeed, letting OPT be the highest score of an attribute combination

$\mathcal{AC} : C \rightarrow \{S \subseteq \mathcal{A} \mid |S| = \ell\}$ and $\mathcal{AC}$ be the one selected by the EM in Algorithm 2, Theorem 3.11 in [17] implies that

$$\Pr \left[\text{GlScore}_\lambda(\mathcal{AC}) \leq \text{OPT} - \frac{2|C|}{\varepsilon} \left(\ln \binom{k}{\ell} + t \right) \right] \leq e^{-t}$$

C Supplementary Experiments

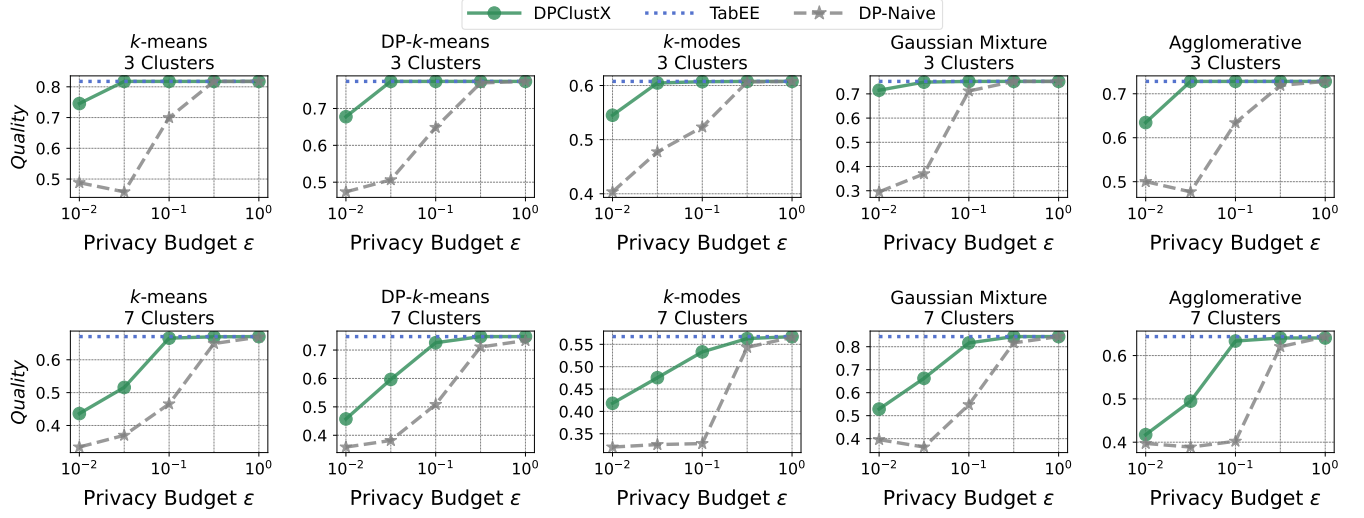
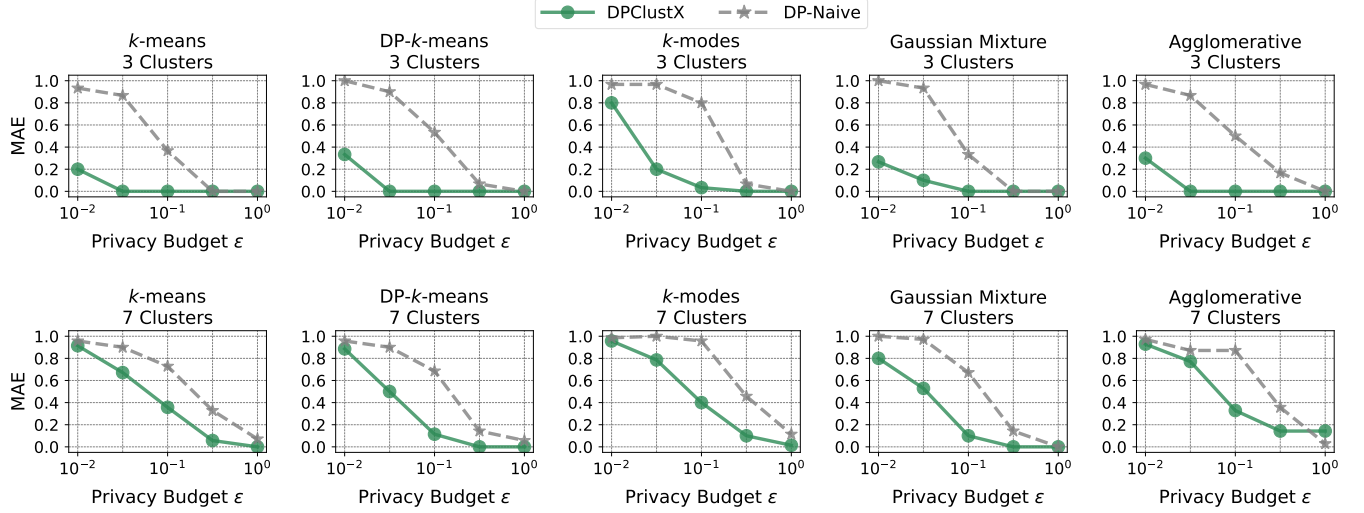
C.1 Selected Attributes Quality score and Error

We present additional results for the Diabetes dataset with 3 and 7 clusters. Figure 10 presents the trend of *Quality* values of the

selected attribute combination as the total privacy budget ε varies. Figure 11 presents the trend of MAE values of the selected attribute combination as the total privacy budget ε varies. Figure 10 contains more results for the diabetes dataset with 3 and 7 clusters.

C.2 Quality for different choices of weights

We present the missing empirical results on the Diabetes (Table 1a), and the Census dataset (Table 1b).

Figure 10: *Quality* values of the selected attribute combination for the Diabetes dataset, as the total privacy budget ϵ varies.Figure 11: *MAE* values of the selected attribute combination for the Diabetes dataset, as the total privacy budget ϵ varies.

Clustering	Explainer	Equal	$\lambda_{\text{Int}} = 0$	$\lambda_{\text{Suf}} = 0$	$\lambda_{\text{Div}} = 0$
<i>k</i> -means	DP-ClustX	0.6874	0.7498	0.7805	0.5319
	TabEE	0.6874	0.7498	0.7805	0.5319
<i>k</i> -modes	DP-ClustX	0.5592	0.6615	0.6821	0.3500
	TabEE	0.5639	0.6641	0.6828	0.3503
Agglomerative	DP-ClustX	0.7255	0.7804	0.7899	0.6061
	TabEE	0.7255	0.7836	0.7899	0.6061
DP- <i>k</i> -means	DP-ClustX	0.7735	0.8263	0.8310	0.6633
	TabEE	0.7735	0.8263	0.8310	0.6633
Gaussian Mixture	DP-ClustX	0.8164	0.8708	0.8518	0.7267
	TabEE	0.8164	0.8708	0.8518	0.7267

(a) Diabetes dataset.

Clustering	Explainer	Equal	$\lambda_{\text{Int}} = 0$	$\lambda_{\text{Suf}} = 0$	$\lambda_{\text{Div}} = 0$
<i>k</i> -means	DP-ClustX	0.8637	0.9195	0.8768	0.7987
	TabEE	0.8643	0.9197	0.8768	0.7987
<i>k</i> -modes	DP-ClustX	0.8247	0.8981	0.8390	0.7449
	TabEE	0.8248	0.8981	0.8390	0.7449
DP- <i>k</i> -means	DP-ClustX	0.8827	0.9552	0.8541	0.8451
	TabEE	0.8828	0.9658	0.8619	0.8451
Gaussian Mixture	DP-ClustX	0.4822	0.6296	0.6009	0.2258
	TabEE	0.4820	0.6296	0.6009	0.2258

(b) Census dataset.

Table 1: *Quality* values for 5 clusters, with different choices of parameters λ_{Int} , λ_{Suf} , and λ_{Div} .